Chapter 2
Energy Balance



2.1 Introduction

The first concept that should be clear when trying to understand energy consumption is the first law of thermodynamics, also called the law of conservation of energy which states that "energy can neither be created nor destroyed; rather, it can only be transformed or transferred from one form to another". This law was first demonstrated by Émilie du Châtelet in 1730. Is a balance of the various forms of energy. One simplified version is expressed as follows

$$\Delta U = Q - W$$

where; ΔU is the change in the thermal energy of the system, Q is the net heat transfer into the system and W is the net work done by the system.

Alternatively;

$$\text{energy input} = \text{energy output} + \text{heat losses}$$

Mass and energy balances are always interrelated, they cannot be separated. The "Handbook on material and energy balance calculations in materials processing" by Morris et al. [1] is a complete reference to learn about mass and energy balances in metallurgy.

In the EAF energy is consumed for heating and melting all the additions. Depending on their physical and chemical composition each material requires a fixed amount of energy that depends on the final temperature of the molten bath. This value is the theoretical energy consumption. Its real value can be higher but not lower. This concept is sometimes misunderstood by some management people in steel plants who demand to decrease energy consumption in order to decrease production costs. The law of conservation of energy in simple words indicates that if we add cheaper raw materials containing impurities with high melting point, the

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theoretical energy requirements will be increased. This value becomes a constant when the mass, temperature and chemical composition of the charged materials is defined:

$$\text{Theoretical energy consumption} = \text{Electrical energy} + \text{Chemical energy} = \text{Constant}$$

The most common unit to measure electric energy consumption in the EAF is in kWh. Watt is a unit of power, which multiplied by time yields energy units. Usually, only the amount of electric energy is recorded during a heat and also usually is the value given as a reference on energy consumption, which is wrong because that is not the total energy consumption. To obtain the total energy consumed, the chemical energy should be calculated, however this is not a common practice. Prakash et al. [2] reported the total energy consumption for different mixtures of metallics, indicating a consumption of electric energy of 480 kWh/ton for scrap and a total energy consumption of 600 kWh/ton. Using 70% DRI the electric energy consumption increased to 500 kWh/ton and the total energy consumption was 700 kWh/ton. In modern practices it is intended to decrease the consumption of electric energy by increasing chemical energy in order to decrease production costs. Madias et al. [3] classified into four ranges of electric energy consumption the EAF's depending on the metallic charge: (1) < 300 kWh/ton; EAF's charged with more than 20% hot metal, (2) 300–400 kwh/ton; EAF's charged with pig iron and high quality scrap, (3) 400–450 kWh/ton; EAF's charged with intermediate quality scrap and hot DRI, and (4) > 450 kWh/ton; EAF's charged with cold DRI or low quality scrap.

The development of energy balances is necessary to define strategies to decrease energy consumption. The following are some of its applications:

- Optimum selection of the charge mixture
- Optimum C–O balance
- Quantify the effect of pre-heating temperature.

There are two types of energy balances, static and dynamic. The static model is easier and faster to develop but is mainly valid for the range of the variables investigated and cannot be extrapolated under different conditions. A dynamic model (also called mechanistic, analytic or phenomenological) is more complex and computationally expensive to develop but can be extrapolated for values outside the range of the variables investigated.

2.2 Components of an Energy Balance

2.2.1 Energy Inputs

The energy sources for the EAF are electric and chemical energy, as shown below.

i. Electric energy
ii. Chemical energy

   a. Chemical energy due to oxygen injection (oxidation reactions)
   b. Chemical energy due to carbon from electrodes
   c. Chemical energy due to organics from scrap
   d. Chemical energy from burners
   e. Chemical energy from post-combustion.

Electric energy: Steel scrap in contrast to hot metal, has a much lower carbon concentration and therefore cannot produce enough chemical energy to melt the metallic charge. The EAF is therefore a suitable reactor to melt steel scrap. Electric energy is still today the main source of energy in conventional EAF's. Electric energy is usually considered a clean energy, however, 62.8% of the electric energy in the world is produced by fossil fuels [4]; 36.4% from coal and 23.3% from natural gas. China which produces almost 30% of the total electricity in the world has the largest consumption of coal for the production of electricity.

Electric energy is supplied from generation plants at high voltages, 100–500 kV. At the steel plant, step-down transformers decrease the voltage to 34.5 kV and finally at the steel shop another transformer decrease the voltage to 100–1000 V and high currents from 40 to 60 kA. The final arc voltage is controlled by the arc length, approx. 10 V/cm.

Chemical energy due to oxygen injection: Oxygen injection is the primary source of chemical energy in the EAF. This is due to the exothermic nature of oxidation reactions. On average 70% of oxygen is consumed by decarburization of the molten metal and the remaining 30% with other alloying elements [5, 6]. The energy input from oxygen injection (kWh/m^3 O_2) depends on the chemical composition and temperature of liquid steel as well as the specific value of oxygen injected. As a reference, the energy input from oxygen injection ranges from 100 to 250 kwh/ton [7].

Chemical energy due to carbon from electrodes: The electrodes as a source of carbon is not desired however it is unavoidable to some extent. Two forms of electrode consumption provide carbon; tip sublimation and side oxidation. Bowman [8, 9] reported expressions indicating that tip consumption is proportional to the square of the electrical current and side oxidation is proportional to the electrode oxidizing surface.

$$E_T\,(\text{kg/ton}) = A\frac{I^2}{P}\frac{t_{on}}{t_{tap}} = A\frac{I^2 t_{on}}{W}$$

$$E_S\,(\text{kg/ton}) = 3B\frac{\pi \overline{D} L_{ox}}{P}$$

$$E_{total} = E_T + E_S$$

where: E_T and E_S are tip and side oxidation in kg/ton, respectively, I is the electric current in kA, $\overline{D}$ is the average electrode diameter in m, L_{ox} is the oxidizing length

of the electrode, W are the tons of liquid steel, t_{on} is the power on time in hr and t_{tap} is the tap-to-tap time in hr. P is the productivity in tons per heat time. A and B are constants, as shown in Table 2.1.

The average diameter can be estimated as follows: $\overline{D} = (D_E + D_T)/2 = (D_E + 0.7D_E)/2$. The oxidizing length, can also be estimated as follows: $L_{ox}(m) = 2.8\sqrt[3]{W/100}$. About 40% is side oxidation and 50% tip oxidation. Electrode consumption in the range from 1.3 to 2.7 kg/ton releases from 12.4 to 24.8 kwh/ton [7]. The heat of formation of CO_2 is about 9.1 kWh/kg C. The oxygen sources for electrode oxidation can be injected oxygen, air infiltration and CO_2 from post-combustion. The oxidation reactions are accelerated due to the high temperature of the surface of electrodes.

Chemical energy due to organics from scrap: Scrap can contain from 0.25 to 1% of oils, plastics and paints which upon combustion releases chemical energy from 25 to 100 kWh/ton [7]. The heat released when some organics are burned is intense and observed in the form of large flames however this source of heat has negative side effects because during combustion polychlorinated dibenzodioxins (PCDD), for short dioxins, and dibenzofurans (PCDF), for short furans, can be formed in concentrations between 0.02 and 9.2 ngTE/m^3 [10], where TE is toxic equivalent. Dioxins and furans are extremely toxic. The low concentrations makes sampling and analysis a complex task [11, 12]. Prüm et al. reported that applying activated HOK lignite before the bag filter house decreased levels from 1.4 to less than 0.1 ngTE/m^3 and comply with EU regulations [10]. Even though dioxins are emitted in relatively low concentrations, they are toxic bio-accumulative compounds [13] and are also one reason to consider EAF dust as a hazardous material because they condense as liquids in the dust collecting system.

Chemical energy from burners and post-combustion: Burners and post-combustion are two important sources of chemical energy in scrap-based operations. They are currently employed to decrease electric energy consumption.

Chapter 4 discusses in detail chemical energy from oxygen injection. Chapters 5 and 6 explains in detail burners and post-combustion.

Table 2.1 Values of parameters A and B, required to estimate electrode consumption [8, 9]

	A, kg/kA^2 h		B, kg/m^2h
AC, arc voltage > 250 V	0.013 × 3	Ladle furnaces	3.0–4.0
Av, tip angle below 30°	0.010 × 3	Closed furnace, < 4 Nm^3O_2/ton	3.0–4.0
AC, tip angle over 40°	0.016 × 3	5–15 Nm^3O_2/ton	5.0–6.0
DC	0.0124	15–45 Nm^3O_2/ton	6.0–8.0
Ladle furnace	0.01–0.03	25–45 Nm^3O_2/ton + post comb	8.0–10.0

2.2.2 Energy Outputs

i. Liquid steel
ii. Liquid slag
iii. Off-gas and dust
iv. Water cooling
v. Heat losses.

Liquid steel: The computation of the theoretical value of the energy required to produce one ton of any material is based on a simple equation that computes the amount of heat at a given temperature.

$$\int_{\Delta H^0_{T_1}}^{\Delta H^0_{T_2}} d\left(\Delta H^0\right) = \int_{T_1}^{T_2} \Delta C_p dT$$

$$\Delta H^0_{T_2} = \Delta H^0_{T_1} + \int_{T_1}^{T_2} \left(\sum \nu_p C_{p,\,\text{products}} - \sum \nu_r C_{p,\,\text{reactants}}\right) dT$$

where: ΔH^0 represents heat or enthalpy, $\Delta H^0_{T_i}$ is the amount of heat at temperature T_i, C_p is the heat capacity, ν_p and ν_r represent the stoichiometric coefficients for products or reactants, respectively.

The energy required to produce one ton of liquid steel at 1537 °C is 337 kWh. This value is for melting pure iron. Additional heat or thermal energy is required for superheating liquid steel and melting non-ferrous materials in the metallic charge. The highest proportion, 76% of the total energy, is required to reach the melting point and the rest is the energy for melting and superheat to 1600 °C.

Liquid slag: The volume of slag produced is defined from the following sources; (i) amount of impurities in the metallic charge, (ii) amount of fluxes added and (iii) oxidation of elements. Scrap and hot metal-based operations produce less slag compared with the use of DRI. The refining capacity of slags increases with its mass, but on the other hand, by increasing the mass of slag also decreases the metallic yield and energy losses. The blast furnace produces 250–300 kg/ton, the BOF produces 100–150 kg/ton and the EAF produces about 100 kg/ton using scrap and 100–150 kg/ton using DRI. In other words, production of ironmaking slags ranges from 25 to 30% and steelmaking slags ranges from 10 to 15% [14]. Therefore, it is to be expected higher energy losses in the slag using DRI in comparison with scrap. Energy losses in the slag in scrap-based operations has been reported in the range from 30 to 50 kWh/tonls [7].

The sensible heat in liquid slag at 1593 °C is 480 kWh/ton slag and increases 0.742 kWh/ton slag for every °C above that temperature [7].

Energy in the off-gas and dust: The rate of production of gases during steelmaking in the EAF is variable during a heat. Increases by increasing the power input, by increasing the oxidation of fossil fuels and by increasing air infiltration.

Energy in the off-gas in scrap-based operations has been reported in the range 80–120 kWh/ton, [5, 15] representing about 10–20% of the input energy.

Energy losses by water cooling: Energy losses in modern EAF's increased due to application of water-cooling systems, however it helped to solve the problem of a long arc operation. Energy losses due to water cooling has been reported in the range 60–100 kWh/ton [7]. Power losses by water cooling include shell, roof, slag door and the fourth hole. The energy in water cooling is defined by the following expression:

$$\Delta H_{wcp} = \left(\sum \dot{m} C_{pw} \Delta T\right) time_{tap\text{-}to\text{-}tap}$$

where: $\dot{m}$ is the water flow rate in water panels of different sections of the EAF, C_{pw} is the heat capacity of water (4180 J/kg °C), ΔT is the temperature gradient between the outlet and inlet temperatures, and $time_{tap\text{-}to\text{-}tap}$ is the tap-to-tap time.

The energy losses from water cooled panels depend on the water flow rate, the construction material, cooling system (water or spray cooling), the thickness of slag layer, energy radiated to the walls and tap-to-tap time.

The energy radiated to the walls increases by increasing the electric power supply especially if the electric arcs are not covered by foamy slag. Simon reported that the structure of the slag layer is formed by alternate layers of slag and metal [16]. Opfermann and Riediger [17] reported that the maximum energy losses occur during the flat bath period, about 1.7 kWh/ton per minute, however they didn't specify the arc coverage. During power-off time they suggest energy losses about 0.5 kWh/ton per minute, equivalent to 7 kWh/ton assuming a total power-off time of 11 min, which includes tapping.

Water cooling is reviewed in detail in Chap. 15.

2.2.3 Energy Losses

Energy losses define the thermal efficiency. In principle, the EAF's thermal efficiency is a ratio between the energy in liquid steel to the energy supplied:

$$\eta_{th} = \frac{Q_{steel}}{Q_{input}}$$

This definition would define energy losses as any energy outlet other than that in liquid steel. This is not the conventional definition for heat losses as explained in the following section. Energy losses is one of the most complex terms in the calculation of an energy balance. It is usually applied as a lump value to make possible the energy

balance by which the energy inlets are equal to the energy outlets. EAF energy losses are influenced by many factors such as [18]:

1. Radiation from the electric arc to the water-cooled panels, which depend on:
   - 1.1 Foaming conditions
   - 1.2 Slag/steel temperature
   - 1.3 Furnace power (furnace transformer) and furnace size
   - 1.4 Tap to tap time
2. Energy losses with slag, which depend on:
   - 2.1 Amount and slag chemical composition
   - 2.2 Energy recovery from liquid slag
3. Energy losses in the off-gas, which depend on:
   - 3.1 Suction power by the fume extraction system
   - 3.2 Volume and chemical composition of the off-gas
   - 3.3 Energy losses due to infiltrated air
   - 3.4 Energy recovery from the off-gas
4. Energy losses due to scrap charging
   - 4.1 Number of scrap charges
   - 4.2 Duration of scrap charges
5. Energy losses due to delays

In order to better understand the term energy losses is necessary to look in detail the general energy balance equation:

$$\text{energy input} = \text{energy output} + \text{heat losses}$$

Energy input has two components; electric (ΔH_e^i) and chemical energy (ΔH_{ch}^i).

$$\text{energy input} = \left(\Delta H_e^i\right) + \left(\Delta H_{ch}^i\right)$$

The chemical energy results from the following chemical reactions:

($\Delta H_{ch,1}^i$) Oxidation reactions with elements dissolved in liquid steel.
($\Delta H_{ch,2}^i$) Oxidation of carbon from the electrodes.
($\Delta H_{ch,3}^i$) Combustion of organic matter in scrap.
($\Delta H_{ch,4}^i$) Combustion of fuel by burners.
($\Delta H_{ch,5}^i$) CO post-combustion.

The energy outputs correspond to the sensible heat in the streams of materials leaving the furnace, endothermic reactions, radiation and convection from different surfaces and energy losses. A total of about 17 terms:

(ΔH_{st}^o) Sensible heat in liquid steel.

(ΔH^{o}_{sl})	Sensible heat in liquid slag.
(ΔH^{o}_{off})	Sensible heat in the off-gas.
(ΔH^{o}_{dust})	Sensible heat in the dust.
(ΔH^{o}_{wc})	Sensible heat in water cooling.
$(\Delta H^{0}_{endo,1})$	Endothermic heat due to reduction of FeO in the slag.
$(\Delta H^{o}_{endo,\,2})$	Endothermic slag's heat of formation.
(ΔH^{o}_{w1})	Endothermic heat to remove humidity in the charged materials.
(ΔH^{o}_{w2})	Endothermic heat to evaporate water cooling of electrodes.
(ΔH^{o}_{foam})	Heat losses due to arc radiation as a function of the foaming conditions.
$(\Delta H^{o}_{power\text{-}off})$	Heat due to radiation and convection from the molten bath during power-off.
(ΔH^{o}_{sch})	Heat due to radiation during scrap charging.
(ΔH^{o}_{sh})	Heat transfer from furnace shell to the surroundings.
(ΔH^{o}_{sd})	Heat transfer from slag door to the surroundings.
(ΔH^{o}_{tap})	Heat transfer from the stream of liquid steel during tapping.
(ΔH^{o}_{hh})	Sensible heat to keep temperature of hot heel.
(ΔH^{o}_{e})	Electric losses.

Taking into account the previous terms, a more detailed description of the heat balance is the following:

$$\begin{aligned}
&\Delta H^{i}_{e} + \left(\Delta H^{i}_{ch,1} + \Delta H^{i}_{ch,2} + \Delta H^{i}_{ch,3} + \Delta H^{i}_{ch,4} + \Delta H^{i}_{ch,5}\right) \\
&= \Delta H^{o}_{st} + \Delta H^{o}_{sl} + \Delta H^{o}_{off} + \Delta H^{o}_{dust} \\
&\quad + \Delta H^{o}_{wc} + \left(\Delta H^{o}_{endo,1} + \Delta H^{o}_{endo,2}\right) + \Delta H^{o}_{w1} \\
&\quad + \Delta H^{o}_{w2} + \Delta H^{o}_{foam} + \Delta H^{o}_{power\text{ - }off} \\
&\quad + \Delta H^{o}_{sch} + \Delta H^{o}_{sh} + \Delta H^{o}_{sd} + \Delta H^{o}_{tap} + \Delta H^{o}_{hh} + \Delta H^{o}_{e}
\end{aligned}$$

If heat losses are all the energy outputs except the sensible heat of liquid steel:

$$\begin{aligned}
\text{heat losses} &= \text{energy input} - \text{sensible heat in liquid steel} \\
\text{heat losses} &= \text{energy input} - \Delta H^{o}_{st}
\end{aligned}$$

The common idea about heat losses is that those terms that are usually more difficult to measure or to estimate and cannot be included in the heat balance are altogether lumped as the heat losses term and therefore result in a very subjective value. Taking into consideration two aspects; first, that it is possible to measure or to estimate the sensible heat in the slag and the sensible heat in the off-gas and, second, the potential for heat recovery from both sources, heat losses can be defined as follows:

$$\begin{aligned}
\text{heat losses} &= \text{energy input} - \left(\Delta H^{o}_{st} + \Delta H^{o}_{sl} + \Delta H^{o}_{off}\right) \\
\text{heat losses} &= \Delta H^{o}_{endo,1} + \Delta H^{o}_{endo,2} + \Delta H^{o}_{dust} \\
&\quad + \Delta H^{o}_{wc} + \Delta H^{o}_{e} \Delta H^{o}_{w1} + \Delta H^{o}_{w2}
\end{aligned}$$

$$+ \Delta H^{o}_{foam} + \Delta H^{o}_{power\ -\ off} + \Delta H^{o}_{sch}$$
$$+ \Delta H^{o}_{sh} + \Delta H^{o}_{sd} + \Delta H^{o}_{tap} + \Delta H^{o}_{hh}$$

Heat losses from the previous equation assumes that only the sensible heat in liquid steel, slag and off-gas is known leaving the rest of the terms as global heat losses, which include all endothermic reactions, arc radiation, delays, scrap charging, furnace shell heat transfer, tapping, energy in the hot heel and electric losses. Water cooling due to its relatively low temperature in comparison with liquid steel, liquid slag and the off-gas is not a source of exergy and its value should be added as part of the heat losses term, however this term can be easily measured and is usually not reported in the heat losses term.

The problem to have an accurate value of heat losses is not only the large number of terms involved which would require a large amount of experimental measurements but also that some parameters are difficult to measure or to estimate. Consequently, the term heat losses is commonly defined in a very loose way. Real heat losses should correspond to heat that leaves in the effluent streams or lost to the atmosphere and is not recovered back into the system. In practice, there is ambiguity in its definition. Heat losses in the EAF should be carefully reported. For example, Diaconu et al. [19] computed an important number of energy losses, reporting that the primary heat losses in a 6.7 ton EAF were due to water cooling (33%), followed by roof opening during scrap charging (29%) and the off-gas (18%), however since many details were not reported, these results are questionable because of inconsistencies in the values reported.

Table 2.2 summarizes a number of heat balances, showing large differences, primarily due to large differences in raw materials and operation condition but very useful to define a characteristic range.

In order to develop an energy balance with higher accuracy it is important to consider the need to start with reliable information. This type of information requires equipment properly calibrated, sensors to monitor the properties indicated, etc. The following list gives an idea of some of the requirements to obtain reliable information.

$\Delta H^{i}_{ch,1}$	Oxidation of elements in metallic charge → accurate amount of each component in the metallic charge and its chemical composition.
(ΔH^{o}_{sl})	Sensible heat in liquid slag → Accurate amount of slag produced and average chemical composition.
(ΔH^{o}_{off})	Sensible heat in the off-gas → Accurate chemical composition of the off-gas, temperature and volume.
$(\Delta H^{i}_{ch,4})$	Combustion of fuel by burners → Accurate value on burners thermal efficiency.
$(\Delta H^{i}_{ch,5})$	CO post-combustion → Accurate value on post-combustion efficiency.
(ΔH^{o}_{dust})	Sensible heat in the dust → Accurate amount of dust produced and chemical composition.
(ΔH^{o}_{wc})	Heat losses in water cooled panels → Inlet and outlet temperature as well as water flow rate for each panel.

Table 2.2 Different energy balances and heat losses reported

		ΔH^{o}_{st}, %	ΔH^{o}_{off}, %	ΔH^{o}_{sl}, %	ΔH^{o}_{wc}, %	ΔH^{o}_{e}, %	Other heat losses, %
1986	Pearce [20]	52.7	19.5	12.2	16.2	1.1	–
1998	Hornby [21]	54	36	7	6	–	1
2002	Pujadas [22]	46	27	7	–	–	–
2002	Kirschen [23]	50.8	10.5	4.0	20.0	1.6	4.2
2002	Kirschen [23]	46.7	15.5	4.6	22.9	4.0	6.3
2005	Pfeifer [6]	53.1	20.7	3.8	13.4	10.0	4.6
2007	Cardenas [15]	46.7	11.0	11.3	–	–	31
2007	Prakash [2]	48.8	10.6	4.9	20.0	11.6	4.0
-	Apfel [nd]	58.5	18.4	11.4	8.1	2.9	–
2009	Kirschen [24]	45–60	15–35	5–10	10–20	–	–
2012	Logar [25]	51.4	15.9	5.5	15.0	4.6	7.6
2014	Pesamosca [26]	54	(20)	11.0	(11)	2.0	–
2015	Thekdi [27]	52	16.7	3.1	7.9	1.7	(0.4)
2016	Yang [28]	53.7	18.9	5.8	3.2	-	-
2020	Diaconu [19]	44	18	7.9	33	2.93	29
2023	Liu [29]	53.9	(25)		8.8	3.21	

$(\Delta H^{i}_{ch,2})$ Oxidation of C from electrodes → how much C from electrodes is oxidized?

$(\Delta H^{i}_{ch,3})$ Combustion of organic matter in scrap → how much organic matter in the scrap?

$\Delta H^{i}_{ch,4}$ Energy from burners → what is the thermal burner's thermal efficiency?

$\Delta H^{i}_{ch,5}$ energy from post-combustion → what fraction of O_2 reacts with CO?

(ΔH^{o}_{foam}) Heat losses due to arc radiation → Coverage of electric arcs (foaming height) and electric power.

$(\Delta H^{0}_{endo,1})$ Reduction of FeO in the slag → how much FeO is reduced?

$(\Delta H^{o}_{endo,\ 2})$ Slag's heat of formation → Accurate thermochemical database and slag amount.

(ΔH^{o}_{dust}) Quantity, temperature and chemical composition of EAF dust per heat.

(ΔH^{o}_{w1}) Heat to remove humidity → Humidity in charged materials.

(ΔH^{o}_{w2}) Heat to evaporate water cooling of electrodes → accurate total volume for water cooling, for each water-cooled panel.

$(\Delta H^{o}_{power\text{-}off})$ Heat losses due to delays → Time due to delays, surface temperature of slag and furnace parts. Part of this heat is transferred to the water-cooled panels.

(ΔH^{o}_{sch}) Heat losses due to radiation during scrap charging → Charging time per scrap basket, number of scrap baskets, shell surface area and temperature, roof surface area and temperature.

(ΔH^{o}_{sh}) Heat losses due to heat transfer from slag door to the surroundings → Shell surface area and temperature.

(ΔH^{o}_{sd}) Heat losses through slag door → Area of slag door, volume of air ingress, time opened during a heat.

(ΔH^{o}_{hh}) Heat losses to keep temperature of hot heel → volume of hot heel and initial temperature.

(ΔH^{o}_{tap}) Heat losses during tapping → Surface and temperature of liquid steel stream, tapping time.

(ΔH^{o}_{e}) Electric losses by different components of the primary and secondary circuit.

Due to the large amount of information it is anticipated an error in any heat balance, its magnitude will depend on the quality of information available from the previous list.

The mass of slag can be estimated from a mass balance for CaO or MgO added in the fluxes, requiring knowledge of the slag and dust chemical composition as well as the amount of additions containing CaO or MgO, and additionally the amount of dust, according with the following expression:

$$W_{flux}X^{flux}_{CaO} + W_{DRI}X^{DRI}_{CaO} = W_{slag}X^{slag}_{CaO} + W_{dust}X^{dust}_{CaO}$$

In addition to oxidation reactions and gangue content in the feed materials, the real slag volume should consider refractory wear. Arzpeyama et al. [30] estimated a value of 3 kg/ton which is added to the previous calculation.

Energy losses depending on furnace capacity: Sims [31] in 1963 suggested the following expression to compute heat losses as a function of furnace capacity:

$$Q = 4.52 \times C^{-0.352}_{furnace}$$

where: Q are the heat losses in kWh/min ton and Ci is the furnace capacity in tons.

Energy losses by conduction and radiation from the crucible: Heat conduction is defined by Fourier's heat law of conduction.

$$\dot{Q} = k\frac{(T_2 - T_2)}{d}$$

where: $\dot{Q}$ is the flux of heat transfer, k is the thermal conductivity, d is the average thickness of the refractory lining, T_2 is the inner temperature, T_1 is the outer temperature.

Thermal or infrared radiation from the outer steel shell is defined by the Stefan-Boltzmann Law.

$$\frac{\dot{Q}}{A} = \sigma\left(T_2^4 - T_1^4\right)$$

where: $\dot{Q}$ is the flux of heat transfer, σ is the Stefan-Boltzmann constant (5.6×10^{-8} W/m^{2}K^4), T_2 is the surface temperature, T_1 is ambient temperature.

The energy losses from the crucible are much smaller in comparison with other heat losses and are therefore usually neglected in the overall energy balances.

Summarizing on the reliability of mass and heat balances, the accuracy between predicted and experimental data on an energy balance depends on [32]:

- Accuracy of mass, chemical composition and temperatures of materials included in the energy balance
- Accuracy to integrate all the chemical reactions taking place
- Accuracy of thermochemical database (heat of reaction, formation and dissolution)
- Assumptions: reaction mechanisms (type and physical state of chemical species).

Frequently is necessary to apply empirical approaches or make assumptions to define the parameters needed for an energy balance. For example, the amount of hot heel and its temperature are usually unknown. A typical value is 10% of the weight of the heat but can be as high as 50% for the Consteel EAF. Adams et al. [33] suggested an efficiency factor of energy conversion from chemical energy of oxidation reactions to the steel melt in the range from 0.7 to 0.8. Abraham and Chen [34] indicated that one of the most challenging aspects to develop an energy balance is to define the fraction of oxygen consumed with all the elements dissolved in liquid steel. To solve the problem, they assigned the following order of reactions:

- Combustion of natural gas
- Oxidation of Al, Si, Mn and Fe
- Oxidation of C–CO
- Oxidation of C–CO_2.

In addition to that, they also considered the following assumptions:

- 3% of the metallic charge is lost due to oxidation
- The slag's melting point was 1573 K
- The specific heat capacity of liquid slag is equal to liquid lime
- The total weight of the slag is twice the weight of the fluxes.

2.2.4 EAF Thermal and Electric Efficiency

Thermal efficiency (η_{th}) can be defined as the ratio of the energy in liquid steel to the total input energy.

$$\eta_{th} = \frac{Q_{steel}}{Q_{input}}$$

Pfeifer et al. [5] provided an alternate definition based on the ratio of energy in both liquid steel and slag to the input energy:

$$\eta_{th}\prime = \frac{Q_{steel} + Q_{slag}}{Q_{input}}$$

This definition, $\eta_{th}\prime$, gives higher values in comparison with η_{th}. Table 2.3 shows a summary of reported thermal efficiencies in EAF's compiled by Pfeifer et al. indicating a range from 50 to 65%. The EAF low thermal efficiency increases the total minimum energy requirements from 350 kWh/ton for pure iron to 560–680 kWh/ton using 100% scrap [35].

Kirschen et al. [24] carried out an analysis of the thermal efficiency in 70 EAF's, as shown in Fig. 2.1. It can be observed the thermal efficiency ranges in general from 40 to 60%. As the thermal efficiency decreases due to higher heat losses, the total energy requirements increase.

Abuluwefa [36] reported a low thermal efficiency of 48% for an EAF with a nominal capacity of 100 ton charged with both scrap and DRI. Most of the energy employed was electric energy, 86%. After meltdown, the thermal efficiency largely depends on the foaming conditions. Ameling [37] described the thermal efficiency in the range from 36 to 100%, depending on the slag foaming conditions.

The instantaneous thermal efficiency results from the instantaneous heat transfer by conduction, convection and radiation among all the materials and phases present in the EAF, this value changes during a heat. In scrap-based operations, the instantaneous thermal efficiency is close to 100% at the initial period of melting because the arc radiation is used to melt the scrap and the off-gas partially preheats the scrap or leaves at low temperatures but once the arc is not shielded by solid scrap, the temperature of the off gas is higher and the electric arc radiates energy to the walls.

Electric efficiency: Electric efficiency depends on the effective transfer of heat radiated from the arc to the solid or liquid steel. During the boring and meltdown period electric efficiency is high, about 88–92%, because the scrap shields the arc [7]. Electric efficiency can be defined as the ratio between the electric arc power (P_{arc}) to the active power (P_A), as follows [38, 39]:

Table 2.3 Thermal efficiency of EAF ($\eta_{th}\prime$) after ref. [5]

Year	Q_{input}	Q_{Elec}	Q_{s+s}	η_{th}, %	References
1982	789	541	503	63.8	Fett et al. Stahl und Eisen, 102, 1982, 461–465
2001	773	497	442	57.2	
1989	704	487	443	59.2	Brod et al., Stahl und Eisen, 109, 1989, 229–238
2001	758	477	449	59.2	Kirschen et al. 59th EAF Conf., 2001, 737–748
2002	807	510	401	49.7	
1990	729	427	429	58.8	Gripenberg et al. Iron and Steel Engineer, 1990,33–37
1998	680	400	440	64.7	Ehle et al. Steel World, 3 (2001) 2, 24–32
1999	810	393	434	53.5	Ehle et al. Steel World, 3 (2001) 2, 24–32

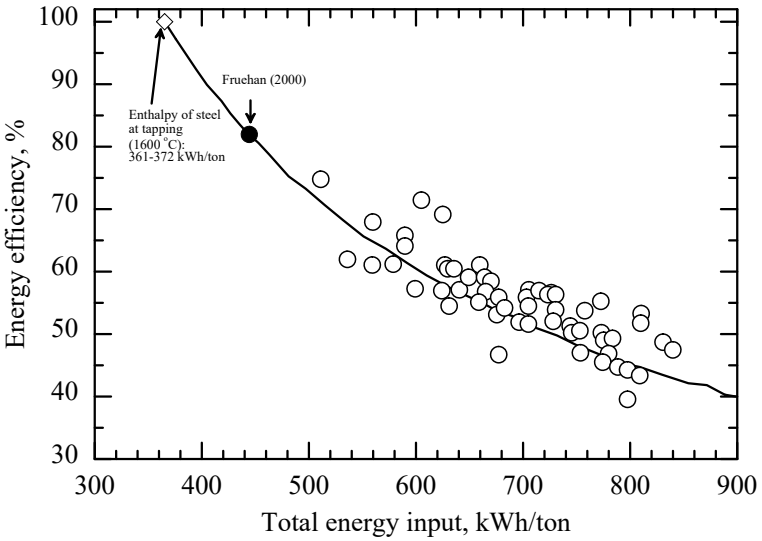


Fig. 2.1 EAF thermal efficiency, after Kirschen et al. [24]

$$\eta_{el} = \frac{P_{arc}}{P_A} = \frac{3\left(\sqrt{U^2 - I^2X^2} - IR\right)I}{3\left(\sqrt{U^2 - I^2X^2} - IR\right)I + 3I^2R} = \frac{3\left(\sqrt{U^2 - I^2X^2} - IR\right)I}{\sqrt{3}UI\cos\emptyset}$$

where: U is voltage, R is electric resistance and cosØ is the power factor.

Kirschen et al. [40] suggest that the electric efficiency is the product of two parts; an efficiency of electric energy transfer to the high current system which ranges from 0.88 to 0.92 and an efficiency of energy conversion from the electric arc which ranges from 0.36 to 0.93, depending on the foaming conditions. The higher value of energy conversion from the electric arc to the molten bath can be obtained with a long arc operation and good foaming conditions (0.88–0.92).

2.2.5 Sankey Diagrams

The metallic charge defines the energy balance. The main raw materials with a higher purity in iron units will result in a lower energy consumption and vice versa. Energy balances are specific to the chemical composition and temperature of the raw materials employed as well as a given set of chemical reactions and heat transfer efficiencies. A review of several cases can give an idea of the magnitude of the total energy inputs and outputs and how this energy is distributed. The graphic representation of an energy balance is given in the form of Sankey diagrams [41].

Table 2.4 EAF energy balance, kWh/ton [35]

Input		Output	
Electric energy	340–500	Liquid steel	350–380
Oxygen–carbon reactions	100–250	Liquid slag	30–50
Burners	25–80	Water cooling	60–100
Scrap grease and others	40–80	Off-gas	140–180
Electrodes	15–25	Recharges and electrical losses	20–80

Pujadas et al. [22] reported an energy balance for a 150 ton AC-EAF with a nominal diameter of 6.7 m, 90 MVA transformer, average active power of 78 MW, charged with 100% scrap, specific oxygen injection of 52 Nm^3/ton and 37.1% FeO, among some of the process conditions. On this basis, the total energy input was 775 kWh/ton, approximately 60% was chemical energy and 40% electric energy. Energy in liquid steel was 46%, in liquid slag 7% and 27% in the off-gas.

Jones et al. [35] reviewed several EAF energy balances from the late 1990s, considering scrap-based operations. For a total energy input of about 560–690 kWh/ton, the input and output energy values are defined in Table 2.4.

Yang et al. [28] reported an energy balance using scrap + 50% hot metal. The electric energy consumption was 23% and 77% was due to chemical energy (24% sensible heat in the hot metal and 34% due to exothermic reactions). For EAF's with intensive post-combustion the energy in the off-gas is about 35–38% [42].

2.3 Static Energy Balances

The theoretical energy consumption is computed using only thermodynamics, which depends only on the initial and final states. A model based only on thermodynamics is a static model.

In the energy balance equation; energy input = energy output + heat losses, the energy input comprises electric and chemical energy and the energy output correspond to the sensible heat in liquid steel, liquid slag, off-gases and energy losses. Energy transfer due to water cooling is part of the energy losses. Energy inputs correspond to the heat required for heating, melting and superheat of the feed materials. The chemical energy is due to oxidation of dissolved elements in liquid steel, reduction of iron oxide in the slag, heats of dissolution of solid carbon and combustion reactions. The main heat losses are due to the heat extracted by the water-cooling systems.

1. The energy required for heating (includes melting and superheat) of individual components in the metallic charge, fluxes, carbon and any other feed material into the EAF is computed with the following expression:

$$\frac{\text{kWh}}{\text{ton}} = \int_{T_1}^{T_2} \sum C_p dT + \Delta H_{transf}$$

where: kWh/ton is the specific energy consumption, C_p is the heat capacity of each material, T_1 and T_2 are the initial and final temperatures, and ΔH_{transf} is the heat of transformation (phase change during heating or melting). Increasing the melting point increases the energy requirements. Carbon plays a key role during steelmaking operations, the melting point of steel scrap is usually higher than that of DRI and pig iron due to differences in carbon concentration, typically about 0.4%, 1% and 4%, respectively. Increasing the carbon concentration decreases the melting point from pure iron, from 1535 to 1153 °C at the eutectic point, corresponding to 4.5%C.

2. Chemical energy for the following chemical reaction is computed using Kirchhoff's law of enthalpy:

$$aA + bB = mM + nN$$

$$\frac{\text{kWh}}{\text{ton}} = \int_{T_1}^{T_2} \left[\left(mC_{p,M} + nC_{p,N} \right) - \left(aC_{p,A} + bC_{p,B} \right) \right]$$

3. Sensible heat in liquid steel, slag, off-gas and water cooling can be calculated with the following equation:

$$\frac{\text{kWh}}{\text{ton}} = \int_{T_1}^{T_2} \sum C_p dT$$

Kirschen et al. [40] used a modified approach. Instead of using mass balances they considered a balance on the rate of mass inputs and outputs which allows to transform the derivatives as a function of time, essentially computing kWH/ton per minute. In the end, they also faced the problem to deal with heat losses or efficiency factors to adjust the heat balance. They reported that in order to reproduce the experimental values they had to adjust locally each one of the efficiency factors, suggesting the following ranges:

$$0.6 < \text{Electric efficiency} < 0.8$$
$$0.7 < \text{Chemical efficiency} < 0.8$$
$$\text{Burner efficiency} < 0.6$$
$$\text{Post-combustion efficiency} < 0.4$$

Using these correction factors these authors studied different cases. One case consisted of an 80-ton EAF charged with 90% DRI + 10% Scrap, the composition

of DRI was 93% metallization (8.3% FeO), 2% C, 2.4% gangue content (1% CaO, 1.4% SiO_2), 57 kg fluxes/ton, injecting 14 kg C/ton and 32 Nm^3 O_2/ton. The amount of slag produced was 11.9%. Their energy balance calculations reported 311 kWh/ton of chemical energy, 582 kWh/ton of electric energy and a total energy demand of 893 kWh/ton.

2.3.1 Variables Affecting Energy Consumption

The total energy consumption in the EAF is not only affected by the amount and type of materials charged into the furnace. A list of the variables that affect energy consumption is given below:

- Quality of raw materials:
  - Steel scrap: density, organic material, residual elements, etc.
  - DRI: Chemical composition (metallization, carbon and gangue concentration) and physical properties (particle size distribution, fraction of fines)
  - Fluxes: amount and type of fluxes
- Metallurgical practice
  - Volume and flow rate of oxygen injected
  - Temperature of liquid steel during melting and tapping
  - Slag foaming practice
  - Amount of slag
  - Scrap charges: number and duration
  - DRI feeding rate
- Process time
  - Tap-to-tap time
  - Power-on time
  - Power-off time
- Heat losses
  - Water cooled panels: flow rate and water temperature, maintenance and slag coating
  - Off-gas extraction system: extraction rate of the off-gases and system's maintenance
  - Slag door opening: Amount of infiltrated air. Heat losses due to air ingress can be in the order of 5 kWh/ton. Closed slag door operation and good maintenance minimize air infiltration
  - Recycling sensible heat in the off-gas and slag
  - Energy radiated from the electric arcs
  - Refractory; refractory life (refractory thickness), corrosion resistance to the slag, erosion rate.

Table 2.5 Theoretical energy requirements to produce steel from 100% scrap with different amounts of slag

Temp., °C	Steel, kwh/ton	Steel + 5% slag, kwh/ton	Steel + 8% slag, kwh/ton	Slag, kwh/ton
1520	368	390	411	508
1594	386	413	428	534
1648	398	426	443	553

- Electric power profile
  - Transformer capacity and electric ancillary equipment for arc stability
  - Electrode regulation
  - Melting rate: Depends not only on the transformer capacity.

The theoretical requirements of energy to melt one ton of scrap is summarized in Table 2.5 as a function of superheat and the amount of slag.

2.3.2 *Energy Consumed to Melt One Ton of Scrap*

This section is a detailed calculation to compute the energy consumed to melt one ton of steel scrap. In order to simplify the energy requirements for scrap melting it is assumed it is pure iron. The heat capacities of iron as a function of temperature (Fe-α, Fe-β, Fe-γ, Fe-δ, Fe-*l*) and heat of transformation are shown in Tables 2.6 and 2.7, respectively.

Table 2.6 Heat capacity of iron as a function of temperature

	°C	°K	C_P, cal/°K mol
Fe-α	25–760	298–1033	$3.0 + 7.58 \times 10^{-3}\ T + 0.6 \times 10^{5}\ T^{-2}$
Fe-β	760–910	1033–1183	11.13
Fe-γ	910–1400	1183–1673	$5.80 + 2.0 \times 10^{-3}\ T$
Fe-δ	1400–1537	1673–1810	$6.74 + 1.64 \times 10^{-3}\ T$
Fe-*l*	1537–2700	1810–2973	$9.77 + 0.40 \times 10^{-3}\ T$

Source O. Kubaschewski and C. Alcock; Metallurgical Thermochemistry, 5th edition, 1989

Table 2.7 Heat of transformation

Phase change	ΔH^0_{transf}, cal/mol
α → β	326
β → γ	215
γ → δ	165
δ → *l*	3670

The amount of heat to increase the temperature of from T_1 to T_2 is given by the following general expression:

$$\Delta H^0_{T_2} = \Delta H^0_{T_1} + \int_{T_1}^{T_2} \sum C_P \, dT$$

If T_2 is 1600 °C (1873 K):

$$\begin{aligned}\Delta H^0_{1873^\circ K} = &\int_{298}^{1033} C_{P,Fe}\alpha \, dT + \Delta H^0_{transf(\alpha\to\beta)} + \int_{1033}^{1183} C_{P,Fe}\beta \, dT \\ &+ \Delta H^0_{transf(\beta\to\gamma)} + \int_{1183}^{1673} C_{P,Fe}\gamma \, dT + \Delta H^0_{transf(\gamma\to\delta)} \\ &+ \int_{1673}^{1810} C_{P,Fe}\delta \, dT + \Delta H^0_{transf(\delta\to l)} + \int_{1810}^{1873} C_{P,Fel} \, dT\end{aligned}$$

Replacing the values of heat capacities:

$$\begin{aligned}\Delta H^0_{1873^\circ K} = &\int_{298}^{1033} \left(3.0 + 7.58 \times 10^{-3} T + 0.6 \times 10^5 T^{-2}\right) dT \\ &+ \Delta H^0_{transf(\alpha\to\beta)} + \int_{1033}^{1183} (11.13\ ) \, dT + \Delta H^0_{transf(\beta\to\gamma)} \\ &+ \int_{1183}^{1673} \left(5.80 + 2.0 \times 10^{-3} T\right) dT + \Delta H^0_{transf(\gamma\to\delta)} \\ &+ \int_{1673}^{1810} \left(6.74 + 1.64 \times 10^{-3} T\right) dT + \Delta H^0_{transf(\delta\to l)} \\ &+ \int_{1810}^{1873} \left(9.77 + 0.40 \times 10^{-3} T\right) dT\end{aligned}$$

$$\begin{aligned}\Delta H^0_{1873^\circ K} = &\ 3.0(1033 - 298) + \frac{7.58 \times 10^{-3}}{2}\left(1033^2 - 298^2\right) \\ &- 0.6 \times 10^5 \left(\frac{1}{1033} - \frac{1}{298}\right) + 326 \\ &+ 11.13(1183 - 1033) + 215 + 5.80(1673 - 1183)\end{aligned}$$

$$+\frac{2.0\times10^{-3}}{2}\left(1673^2-1183^2\right)+165$$
$$+6.74(1810-1673)+\frac{1.64\times10^{-3}}{2}\left(1810^2-1673^2\right)+3670$$
$$+9.77(1873-1810)+\frac{0.40\times10^{-3}}{2}\left(1873^2-1810^2\right)$$
$$\Delta H^0_{1873^\circ K}=5993+326+1669.6+215+4241.4$$
$$+165+1314.6+3670+661.91=18,256.9\ \text{cal/mol}$$
$$\Delta H^0_{1873^\circ K}=18{,}256.9\,\frac{\text{cal}}{\text{gmol}}\times\frac{1\,\text{gmol Fe}}{55.854\,\text{g}}$$
$$\times\frac{1000\ \text{g}}{\text{kg}}=326{,}868\,\frac{\text{cal}}{\text{kg Fe}^\circ}$$
$$=326.868\frac{\text{kcal}}{\text{kg scrap}}\times\frac{1000\,\text{kg}}{\text{ton}}\times\frac{\text{kWh}}{860\,\text{kcal}}$$
$$=380\frac{\text{kWh}}{\text{ton scrap}}=1.368\,\frac{\text{GJ}}{\text{ton scrap}}$$

From the total amount of energy required to produce a ton of liquid steel from steel scrap, about 76% is employed to pre-heat the charged materials, 20% for melting and 4% to reach the final molten steel temperature (1600 °C in this case) as illustrated in Fig. 2.2. This information indicates that pre-heating consumes the largest amount of energy.

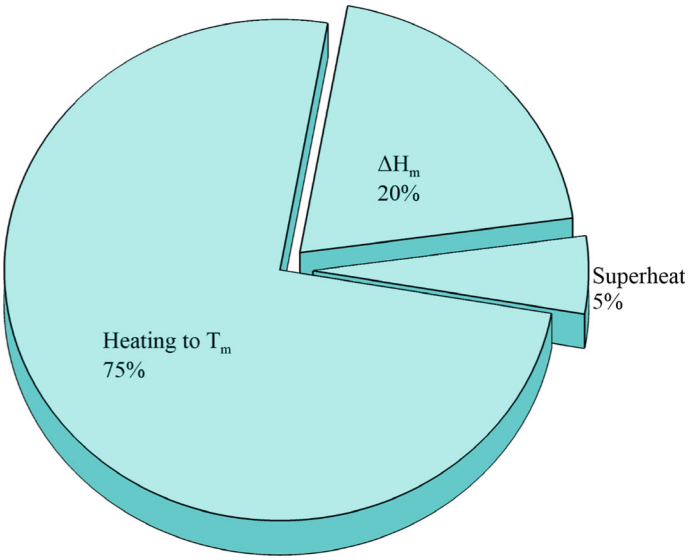


Fig. 2.2 Energy requirements to melt one ton of scrap up to 1600 °C

2.3.3 Energy Consumed to Melt One Ton of DRI

Direct Reduced Iron (DRI) in addition to metallic iron, also contains unreduced oxides of high melting point. The energy requirements to melt DRI are defined considering the specific energy demand for each component and its mass fraction. In addition to metallic iron, DRI also contains the following chemical species: Fe_3C, C, FeO, SiO_2, CaO, Al_2O_3 and MgO. Silica has three polymorphs; Quartz is stable up to 867 °C, tridymite is stable from 867 to 1470 °C and cristobalite from 1470 °C to the melting point of silica. Table 2.8 shows the heat capacities and Table 2.9 the heats of transformation (includes melting).

The following is a detailed calculation on the energy demand to bring all the previous chemical species up to 1600 °C (1873 K):

SiO_2:

$$\Delta H^0_{1873°K} = \int_{298}^{523} C_{P,SiO_2-\alpha}\, dT + \Delta H^0_{transf(\alpha\rightarrow\beta)} + \int_{523}^{1873} C_{P,SiO_2-\beta}\, dT$$

Table 2.8 Heat capacities

Phase	°C	K	C_P, cal/K mol	Source
α-cristobalite	25–250	298–523	$4.28 + 21.06 \times 10^{-3}$ T	1
β-cristobalite	250–1722	523–1995	$17.39 + 0.31 \times 10^{-3}$ T – 9.90×10^5 T^{-2}	1
C	25–2027	298–2300	$4.10 + 1.02 \times 10^{-3}$ T –2.10×10^5 T^{-2}	2
CaO	25–1600	298–1800	$11.67 + 1.08 \times 10^{-3}$ T –1.56×10^5 T^{-2}	2
MgO	25–1827	298–2100	$10.18 + 1.74 \times 10^{-3}$ T –1.48×10^5 T^{-2}	2
Al_2O_3	25–1600	298–1800	$27.43 + 3.06 \times 10^{-3}$ T –8.47×10^5 T^{-2}	2
$Fe_{0.95}O$	25–1377	298–1650	$11.66 + 2.00 \times 10^{-3}$ T – 0.67×10^5 T^{-2}	2
$Fe_{0.95}O$	1377–1600	1650–1873	16.30	2

[a] O. Kubaschewski and C. Alcock; Metallurgical Thermochemistry, 5th edition, 1989
[b] Handbook on Material and Energy Balance calculations in Metallurgical Process Gordon H. Geiger Revised edition, 1993

Table 2.9 Heats of transformation

Phase	Transf	T, K	ΔH^0_{transf}, cal/mol
SiO_2	$\alpha \rightarrow \beta$	523	200
SiO_2	$\beta \rightarrow l$	1986	3100
FeO	$s \rightarrow l$	1650	7490

[a] J. Elliot & M. Gleiser; Thermochemistry for steelmaking, vol I, 1960
[b] Handbook on Material and Energy Balance calculations in Metallurgical Process. G. H. Geiger. Revised edition, 1993

$$\Delta H^0_{1873^\circ K} = \int_{298}^{523} \left(4.28 + 21.06 \times 10^{-3}\,T\right) dT + \Delta H^0_{transf(\alpha\rightarrow\beta)} + \int_{523}^{1873} \left(17.39 + 0.31 \times 10^{-3}\,T - 9.90 \times 10^5\,T^{-2}\right) dT$$

$$\Delta H^0_{1873^\circ K} = 4.28(523 - 298) + \frac{21.06 \times 10^{-3}}{2}\left(523^2 - 298^2\right) + 200 + 17.39(1873 - 523) + \frac{0.31 \times 10^{-3}}{2}\left(1873^2 - 523^2\right) + 9.90 \times 10^5\left(\frac{1}{1873} - \frac{1}{523}\right)$$

$$\Delta H^0_{1873^\circ K} = (963 + 1945) + 200 + (23476.5 + 501 - 1364) = 25,721.5 \text{ cal/mol}$$

$$\Delta H^0_{1873^\circ K} = 25,721.5\ \frac{\text{cal}}{\text{gmol}} \times \frac{1\text{ gmol } SiO_2}{60.0843\text{ g}} \times \frac{1000\text{ g}}{\text{kg}} = 428,090.2\ \frac{\text{cal}}{\text{kg } SiO_2}$$

$$= 428\frac{\text{kcal}}{\text{kg } SiO_2} \times \frac{1000\text{ kg}}{\text{ton}} \times \frac{\text{kwh}}{860\text{ kcal}} = 497.8\frac{\text{kwh}}{\text{ton } SiO_2}$$

C:

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} C_{P,C}\, dT$$

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} \left(4.10 + 1.02 \times 10^{-3}\,T - 2.10 \times 10^5\,T^{-2}\right) dT$$

$$\Delta H^0_{1873^\circ K} = 4.10(1873 - 298) + \frac{1.02 \times 10^{-3}}{2}\left(1873^2 - 298^2\right) + 2.10 \times 10^5\left(\frac{1}{1873} - \frac{1}{298}\right)$$

$$\Delta H^0_{1873^\circ K} = (6457.5 + 1743.85575) - (592.57) = 7608.77 \text{ cal/mol}$$

$$\Delta H^0_{1873^\circ K} = 7608.77\ \frac{\text{cal}}{\text{gmol}} \times \frac{1\text{ gmol C}}{12.0\text{ g}} \times \frac{1000\text{ g}}{\text{kg}} = 634{,}064\ \frac{\text{cal}}{\text{kgC}}$$

$$= 634\frac{\text{kcal}}{\text{kg C}} \times \frac{1000\text{ kg}}{\text{ton}} \times \frac{\text{kwh}}{860\text{ kcal}} = 737\frac{\text{kwh}}{\text{ton C}}$$

CaO:

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} C_{P,CaO}\, dT$$

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} \left(11.67 + 1.08 \times 10^{-3} T - 1.56 \times 10^5 T^{-2}\right) dT$$

$$\Delta H^0_{1873^\circ K} = 11.67(1873 - 298) + \frac{1.08 \times 10^{-3}}{2}\left(1873^2 - 298^2\right) + 1.56 \times 10^5 \left(\frac{1}{1873} - \frac{1}{298}\right)$$

$$\Delta H^0_{1873^\circ K} = (18380.25 + 1846.4355) - (440.20)$$

$$= 19,786.5 \frac{\text{cal}}{\text{mol CaO}}$$

$$\Delta H^0_{1873^\circ K} = 19,768 \frac{\text{cal}}{\text{gmol}} \times \frac{1 \text{ gmol CaO}}{56.07 \text{ g}} \times \frac{1000 \text{ g}}{\text{kg}}$$

$$= 352,888 \frac{\text{cal}}{\text{kg CaO}}$$

$$= 352.8 \frac{\text{kcal}}{\text{kg CaO}} \times \frac{1000 \text{ kg}}{\text{ton}} \times \frac{\text{kwh}}{860 \text{ kcal}} = 410 \frac{\text{kwh}}{\text{ton CaO}}$$

MgO:

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} C_{P,MgO}\, dT$$

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} \left(10.18 + 1.74 \times 10^{-3}\, T - 1.48 \times 10^5\, T^{-2}\right) dT$$

$$\Delta H^0_{1873^\circ K} = 10.18(1873 - 298) + \frac{1.74 \times 10^{-3}}{2}\left(1873^2 - 298^2\right) + 1.48 \times 10^5 \left(\frac{1}{1873} - \frac{1}{298}\right)$$

$$\Delta H^0_{1873^\circ K} = (16,033.5 + 2974.81275) - (417.62)$$

$$= 18,590 \text{ cal/mol}$$

$$\Delta H^0_{1873^\circ K} = 18,590 \frac{\text{cal}}{\text{gmol}} \times \frac{1 \text{ gmol MgO}}{40.302 \text{ g}} \times \frac{1000 \text{ g}}{\text{kg}}$$

$$= 461,284 \frac{\text{cal}}{\text{kg MgO}}$$

$$= 461 \frac{\text{kcal}}{\text{kg MgO}} \times \frac{1000 \text{ kg}}{\text{ton}} \times \frac{\text{kWh}}{860 \text{ kcal}} = 536 \frac{\text{kWh}}{\text{ton MgO}}$$

Al_2O_3:

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} C_{P,Al_2O_3}\, dT$$

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1873} \left(27.43 + 3.06 \times 10^{-3}\, T - 8.47 \times 10^5\, T^{-2}\right) dT$$

$$\Delta H^0_{1873^\circ K} = 27.43(1873 - 298) + \frac{3.06 \times 10^{-3}}{2}\left(1873^2 - 298^2\right) + 8.47 \times 10^5 \left(\frac{1}{1873} - \frac{1}{298}\right)$$

$$\Delta H^0_{1873^\circ K} = (43,202.25 + 5231.56725) - (2390.066) = 46{,}043 \text{ cal/mol}$$

$$\Delta H^0_{1873^\circ K} = 46{,}043\ \frac{\text{cal}}{\text{gmol}} \times \frac{1 \text{ gmol } Al_2O_3}{101.933 \text{ g}} \times \frac{1000 \text{ g}}{\text{kg}} = 451{,}706\ \frac{\text{cal}}{\text{kg } Al_2O_3}$$

$$= 451.7\ \frac{\text{kcal}}{\text{kg } Al_2O_3} \times \frac{1000 \text{ kg}}{\text{ton}} \times \frac{\text{kWh}}{860 \text{ kcal}}$$

$$= 525.23\ \frac{\text{kWh}}{\text{ton } Al_2O_3}$$

$Fe_{0.95}O$:

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1650} C_{P,Fe_{0.95}O}\, dT + \Delta H^0_{\text{transf}(s \to l)} + \int_{1650}^{1873} C_{P,Fe_{0.95}O}\, dT$$

$$\Delta H^0_{1873^\circ K} = \int_{298}^{1650} \left(11.66 + 2.00 \times 10^{-3}\, T - 0.67 \times 10^5\, T^{-2}\right) dT + \Delta H^0_{\text{transf}(s \to l)} + \int_{1650}^{1873} (16.30)\, dT$$

$$\Delta H^0_{1873^\circ K} = 11.66(1650 - 298) + \frac{2.00 \times 10^{-3}}{2}\left(1650^2 - 298^2\right) + 0.67 \times 10^5 \left(\frac{1}{1650} - \frac{1}{298}\right) + 7490 + 16.30(1873 - 1650)$$

$$\Delta H^0_{1873^\circ K} = (15,764.32 + 2633.696 - 184.22)$$

$$+ 7490 + (3634.9) = 29,522.9 \text{ cal/mol}$$

$$\Delta H^{0}_{1873^{\circ}K} = 29,522.9 \frac{\text{cal}}{\text{gmol}} \times \frac{1 \text{ gmol } Fe_{0.95}O}{71.837 \text{ g}} \times \frac{1000 \text{ g}}{\text{kg}}$$

$$= 410,970 \frac{\text{cal}}{\text{kg } Fe_{0.95}O}$$

$$= 410.97 \frac{\text{kcal}}{\text{kg } Fe_{0.95}O} \times \frac{1000 \text{ kg}}{\text{ton}} \times \frac{\text{kWh}}{860 \text{ kcal}}$$

$$= 477.9 \frac{\text{kWh}}{\text{ton } Fe_{0.95}O}$$

Based on the previous calculations, Table 2.10 summarizes the specific energy consumption for the different components in DRI. In the literature there are different values, all having very small differences, due to different values for the heat capacities and heats of transformation, for example Fruehan et al. [43] reported a value of 368 kWh/ton for scrap melting up to 1600 °C, a difference of about 3% with respect to the present calculation.

Heat of formation of slag: Due to the high melting temperature of oxides, at 1600 °C only the melting temperature for pure FeO is reached but not for the other pure oxides. Obviously at 1600 °C the slag is formed and composed of all those oxides because, on one side, its real chemical composition is not in the form of pure oxides but complex compounds that have lower melting points and in addition to this, some high melting point oxides are melted at the high temperatures of the arc zone. A detailed analysis is out of the scope in this section, in order to estimate the heat of formation of the slag, as a first approximation it is assumed only liquid FeO is formed then due to the presence of carbon, its reduction, which is highly endothermic, should also be accounted for. With this approximation, the heat of formation of slag is defined by the reduction of FeO by dissolved carbon:

$$(FeO) + \underline{C} = \underline{Fe} + CO_{(g)}$$

The heat of reaction at 1600 °C was estimated as 651 kWh/ton FeO [44].

The previous values can be employed to get a first estimate of the energy requirements to melt one ton of DRI. As an example, consider a DRI with the following chemical composition:

Table 2.10 Energy requirements to raise the temperature of DRI components to 1600 °C

	SiO_2	CaO	MgO	Al_2O_3	$Fe_{0.95}O$	C	Fe
cal/kg	428,090	352,888	461,284	451,706	410,970	634,064	326,868
MJ/ton	1791	1476	1930	1889	1719	2652	1367
kWh/ton	497.8	410	536	525.23	477.9	737	380

	Fe_t	Fe°	C	SiO_2	CaO	Al_2O_3	MgO	FeO
X_i (wt%)	90.8	85.8	2.48	2.41	0.61	0.87	0.69	6.43

$$\%FeO_{DRI} = (\%Fe_t - \%Fe^\circ)\frac{FeO}{Fe} = (90.8 - 85.8)\frac{72}{56} = 6.43$$

	SiO_2	CaO	MgO	Al_2O_3	$Fe_{0.95}O$	C	Fe
Cal/kg	428,090	352,888	461,284	451,706	410,970	634,064	326,868
MJ/ton	1791	1476	1930	1889	1719	2652	1367
kWh/ton	497.8	410	536	525.23	477.9	737	380

$$\begin{aligned}\Delta H_{DRI}^{1600}\left(\frac{kWh}{ton\,DRI}\right) &= \left(X_{Fe_\circ} \times \Delta H_{Fe}^\circ\right) + \left(X_C \times \Delta H_C^\circ\right) + \left(X_{SiO_2} \times \Delta H_{SiO_2}^\circ\right) \\ &+ \left(X_{CaO} \times \Delta H_{CaO}^\circ\right) \\ &+ \left(X_{Al_2O_3} \times \Delta H_{Al_2O_3}^\circ\right) + \left(X_{MgO} \times \Delta H_{MgO}^\circ\right) \\ &+ \left(X_{Fe_{0.95}O} \times \Delta H_{Fe_{0.95}O}^\circ\right) + \left(X_{Slag} \times \Delta H_{Slag}^\circ\right)\end{aligned}$$

$$\begin{aligned}\Delta H_{DRI}^{1600}\left(\frac{kWh}{ton\,DRI}\right) &= (0.858 \times 380) + (0.0248 \times 737) \\ &+ (0.0241 \times 497) + (0.0061 \times 410) \\ &+ (0.0087 \times 525) + (0.0069 \times 536) \\ &+ (0.0643 \times 477) + (0.0643 \times 651) \\ &= 326 + 18.27 + 11.97 + 2.5 + 4.56 \\ &+ 3.71 + 30.67 + 41.85 = 439.6\,kWh/ton\,DRI\end{aligned}$$

In order to define the energy consumption on terms of per ton of steel, the following experimental conversion factor is employed:

$$1\,ton\,steel = 1.16\,ton\,DRI$$

$$\Delta H_{DRI}^{1600} = 1.16 \times 439.6 = 509\,\frac{kWh}{ton\,steel} = 1.83\,\frac{GJ}{ton\,steel}$$

2.3.4 Chemical Energy Sources

As mentioned before, the chemical energy derives from the following sources: oxidation reactions with elements contained in the metallic charge, combustion from burners, post-combustion, burning of organics in the scrap, electrode oxidation, dissolution of chemical species in liquid steel or liquid slag, calcination of limestone, etc. The analysis of chemical reactions is quite complex because involves not only the number of chemical reactions but the definition of the fraction of oxygen that reacts in a given chemical reaction, the accuracy to define the state of the reacting phases present, etc. The following list is an example of the chemical reactions to be considered in order to estimate the amount of chemical energy:

1. Heat released due to oxidation of Si: $Si + O_2 \rightarrow SiO_2$
2. Heat released due to oxidation of C to CO: $C + ½O_2 \rightarrow CO$
3. Heat released due to oxidation of Fe: $Fe + ½O_2 \rightarrow FeO$
4. Heat absorbed due to the reduction of FeO by carbon dissolved in liquid steel: $FeO + \underline{C} \rightarrow Fe + CO$
5. Reduction of FeO by silicon in scrap: $2FeO + \underline{Si} \rightarrow 2Fe + SiO_2$
6. Heat released by oxyfuel burners: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
7. Heat released due to post-combustion of CO: $CO + ½O_2 \rightarrow CO_2$
8. Heat released by organic matter in the scrap: $C_xH_y + (x + y/4)\ O_2 \rightarrow xCO_2 + \frac{y}{2}\ H_2O$
9. Heat of dissolution of FeO contained in DRI: $FeO_{DRI} \rightarrow \underline{Fe} + ½O_2$
10. Heat of dissolution of Fe_3C in DRI: $Fe_3C \rightarrow 3\underline{Fe} + C$
11. Heat of formation of liquid slag.

In order to define the fraction of oxygen in each of the previous reactions, Abraham and Chen [34] developed an empirical approach, which was adjusted comparing model predictions and experimental data, defining the following priority in the consumption of oxygen:

- Combustion reactions by oxy-fuel burners
- Oxidation of dissolved elements in steel; Al, Si, Mn and Fe
- Oxidation of carbon dissolved in liquid steel to CO
- Post-combustion from CO to CO_2.

The heat of reaction can be estimated in several ways. One way is to let the reactions happen at room temperature and then raise the temperature of the products to the molten bath temperature. Another way is to raise the reactants to the molten bath temperature and then let the chemical reactions occur at that temperature. Figure 2.3 shows the second option.

The thermodynamic databases to carry out the calculations are shown in Tables 2.11, 2.12 and 2.13. Table 2.11 shows the heat capacities, Table 2.12 defines the heats of transformation and melting, Table 2.13 defines the heats of reactions at room temperature.

The heat of reaction at any temperature is computed using Kirchoff's law:

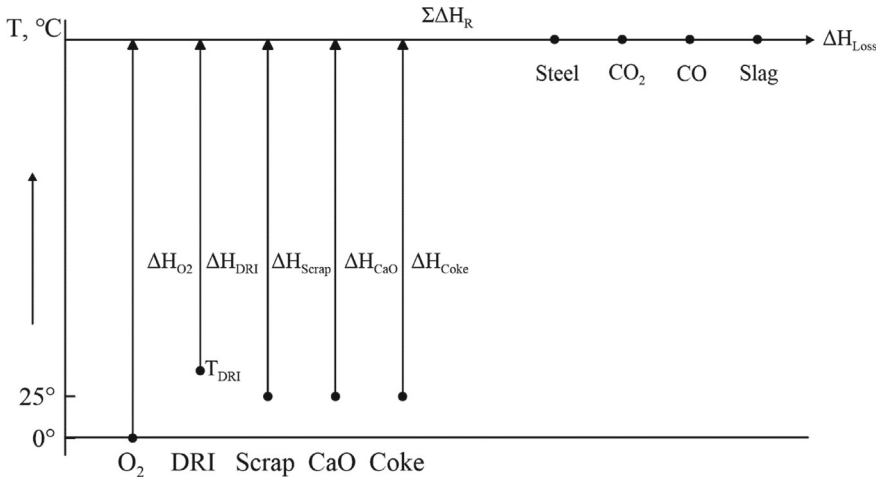


Fig. 2.3 Thermodynamic analysis of the heat of reaction

$$\Delta H^{o}_{R,T} = \Delta H^{o}_{R,298} + \int_{298}^{T} \Delta C_p dT$$

The following computation of heats of reactions is based on the work of Cardenas et al. [15, 44, 45].

(1) $FeO_{DRI} + \underline{C} \rightarrow Fe + CO$

$$\begin{aligned}\Delta H^{o}_{R} = \Delta H^{o}_{R,298} &+ \int_{298}^{1000} \Delta C_p\, dT + \int_{1000}^{1033} \Delta C_p\, dT \\ &+ \int_{1033}^{1179} \Delta C_p\, dT + \int_{1179}^{1644} \Delta C_p\, dT + \int_{1644}^{1674} \Delta C_p\, dT \\ &+ \int_{1674}^{1700} \Delta C_p\, dT + \int_{1700}^{1803} \Delta C_p\, dT \\ &+ \int_{1803}^{T} \Delta C_p\, dT + \Delta H_{Fe,\alpha-\beta} + \Delta H_{Fe,\beta-\gamma} \\ &+ \Delta H_{Fe,\gamma-\delta} + \Delta H^{m}_{Fe,\delta-l} + \Delta H^{m}_{FeO} + \Delta H^{d}_{C}\end{aligned}$$

where:

Table 2.11 Heat capacities

Specie	State	C_p, cal/K mol	Range, K
Fe	c, α	$3.37 + 7.10 \times 10^{-3}\mathrm{T} + 0.43 \times 10^{5}\mathrm{T}^{-2}$	298–1033[a]
	c, β	10.40	1033–1179[a]
	c, γ	$4.85 + 3.00 \times 10^{-3}\mathrm{T}$	1179–1674[a]
	c, δ	10.30	1674–1803[a]
	l	10.00	1803–1900[a]
FeO	c	$-4.307 + 7.315 \times 10^{-3}\,\mathrm{T} - 605,473.23\,\mathrm{T}^{-2} + 358.7237\,\mathrm{T}^{-0.5}$	298–1644[b]
	l	16.3	1644–2000[b]
Fe_3C	c, α	$19.64 + 20.00 \times 10^{-3}\mathrm{T}$	298–463[b]
	c, β	$25.62 + 3.00 \times 10^{-3}\mathrm{T}$	463–1500[b]
	l	29.0	1500–2000[b]
SiO_2	c_{crist}, α	$19.960 - 586,845.12\,\mathrm{T}^{-2} - 89.553\,\mathrm{T}^{-0.5} + 6,693,886\,\mathrm{T}^{-3}$	298–1996[b]
Al_2O_3	c	$-3.760 + 7.157 \times 10^{-3}\mathrm{T} + 1,193,897\,\mathrm{T}^{-2} + 1689.376\,\mathrm{T}^{-0.5} - 26,794\,\mathrm{T}^{-1}$	298–1200[b]
	c	$-188.22 + 15.745 \times 10^{-3}\,\mathrm{T} + 63,053,626\,\mathrm{T}^{-2} + 159,86.07\,\mathrm{T}^{-0.5} - 364,609.8\,\mathrm{T}^{-1}$	1200–2327[b]
CaO	c	$14.051 - 274,174.47\,\mathrm{T}^{-2} - 32.003\,\mathrm{T}^{-0.5} + 24,612,521\,\mathrm{T}^{-3}$	298–2845[b]
MgO	c	$14.605 - 148,459.37\,\mathrm{T}^{-2} - 283.173\,\mathrm{T}^{-0.5} + 1{,}396{,}895.8\,\mathrm{T}^{-3}$	298–2845[b]
S	c_{rom}	$4.126 + 4.781 \times 10^{-3}\,\mathrm{T}\,36.217\,\mathrm{T}^{-1}$	298–368[b]
	c_{mon}	$4.189 + 4.7002 \times 10^{-3}\mathrm{T}$	368–388[b]
	l	$-12.637 + 51.649 \times 10^{-3}\mathrm{T}$	388–432[b]
	l	$13.461 - 8.765 \times 10^{-3}\mathrm{T}$	432–500[b]
	l	$13.503 - 8.843 \times 10^{-3}\mathrm{T}$	500–600[b]
	l	$10.062 - 3.107 \times 10^{-3}\mathrm{T}$	600–800[b]
	l	7.565	800–1500[b]
C	c	$14.622 - 1.916 \times 10^{-3}\,\mathrm{T} + 197{,}315.9\,\mathrm{T}^{-2} - 244.74\,\mathrm{T}^{-0.5} - 1{,}578{,}579\,\mathrm{T}^{-3}$	298–1000[b]
	c	$5.922 + 0.2291 \times 10^{-3}\,\mathrm{T} - 689,151.12\,\mathrm{T}^{-2} - 9.486\,\mathrm{T}^{-0.5}$	1000–6000[b]
CO	g	$21.69 - 1.49 \times 10^{-3}\,\mathrm{T} - 368,368\,\mathrm{T}^{-2} + 7663.3163\,\mathrm{T}^{-1} - 618.846\,\mathrm{T}^{-0.5}$	298–1700[b]
	g	$10.536 - 2,780,374.386\,\mathrm{T}^{-2} + 4993.763\,\mathrm{T}^{-1} - 164.263\,\mathrm{T}^{-0.5}$	1700–6000[b]

(continued)

Table 2.11 (continued)

Specie	State	C_p, cal/K mol	Range, K
Si	*c*	$5.451 + 0.925 \times 10^{-3}\,T - 84367.12\,T^{-2}$	298–1685[b]
	c	6.5	1685–2500[b]
Mn	*c*, α	$5.7002 + 3.379 \times 10^{-3}\,T - 37,536.83\,T^{-2}$	298–980[b]
	c, β	$8.183 + 0.904 \times 10^{-3}\,T - 82,037.66\,T^{-2}$	980–1361[b]
	c, γ	$7.5979 + 1.985 \times 10^{-3}\,T$	1361–1412[b]
	c, δ	$8.0221 + 1.9759 \times 10^{-3}\,T$	1412–1519[b]
	l	11.00	1519–3000[b]
O_2	*g*	$6.435 + 4.058 \times 10^{-3}\,T + 54,811\,T^{-2} - 1.617 \times 10^{-6}\,T^2 - 18.92\,T^{-0.5}$	298–1000[b]
	g	$21.434 - 0.3459 \times 10^{-3}\,T - 4,465,269\,T^{-2} - 986.26^{-0.5} + 22,897.69\,T^{-1}$	1000–4000[b]

Table 2.12 Heat of transformation and melting

Reaction	ΔH_{Trans}, cal/mol	ΔH_{Fus}, cal/mol	T, K
$Fe_{c,\alpha} \rightarrow Fe_{c,\beta}$	410	–	1033[a]
$Fe_{c,\beta} \rightarrow Fe_{c,\gamma}$	210	–	1179[a]
$Fe_{c,\gamma} \rightarrow Fe_{c,\delta}$	110	–	1674[a]
$Fe_{c,\delta} \rightarrow Fe_l$	–	3700	1803[a]
$FeO_c \rightarrow FeO_l$	–	7490	1644[b]
$Fe_3C_{c,\alpha} \rightarrow Fe_3C_{c,\beta}$	180	–	463[b]
$Fe_3C_{c,\beta} \rightarrow Fe_3C_l$	–	12,300	1500[b]
$S_{c,rom} \rightarrow S_{c,mon}$	95.690	–	368.3[b]
$S_{c,mc} \rightarrow S_l$	–	411.328	388.4[b]
$Si_c \rightarrow Si_l$	–	11,998	1685[b]
$Mn_{c,\alpha} \rightarrow Mn_{c,\beta}$	532.02	–	980[b]
$Mn_{c,\beta} \rightarrow Mn_{c,\gamma}$	507.17	–	1361[b]
$Mn_{c,\gamma} \rightarrow Mn_{c,\delta}$	449.09	–	1412[b]
$Mn_{c,\delta} \rightarrow Mn_l$	–	2881.93	1519[b]

[a] G. H. Geiger; Handbook on Material and Energy Balance Calculations in Metallurgical Process, Revised Edition, 1993
[b] FactSage 5.2 Pelton

$$\Delta C_p = C_{p,Fe} + C_{p,CO} - C_{p,FeO} - C_{p,C}$$

Replacing the heat capacities for each temperature range:

Table 2.13 Heat of reaction at 298 K

Reaction	$\Delta H^{o}_{R,\,298}\ \frac{cal}{mol}$
$FeO_{DRI} + \underline{C} \rightarrow Fe + CO$	37,118.84
$Fe_3C_{DRI} \rightarrow 3Fe + \underline{C}$	− 5400.00
$2FeO + \underline{Si} \rightarrow 2Fe + SiO_2$	− 89,558.49
$\underline{Si} + O_2 \rightarrow SiO_2$	− 216,619.35
$\underline{C} + \frac{1}{2}O_2 \rightarrow CO$	− 26,416.58
$Fe + \frac{1}{2}O_2 \rightarrow FeO$	− 63.535.43
$C_{(gr)} \rightarrow \underline{C}$	5100.00
$Si_l \rightarrow \underline{Si}$	31,100.00

$$
\begin{aligned}
\Delta H^{o}_{R} = {} & \Delta H^{o}_{R,\,298} \\
& + \int_{298}^{1000} \left(14.74 + 0.21 \times 10^{-3}T + 82{,}788T^{-2} - 732T^{-0.5} + 7663T^{-1} + 1{,}578{,}578T^{-3}\right) dT \\
& + \int_{1000}^{1033} \left(23.44 - 1.93 \times 10^{-3}T + 969{,}255T^{-2} - 968T^{-0.5} + 7663T^{-1}\right) dT \\
& + \int_{1033}^{1179} \left(30.47 - 9.03 \times 10^{-3}T + 926{,}255T^{-2} - 968T^{-0.5} + 7663T^{-1}\right) dT \\
& + \int_{1179}^{1644} \left(24.92 - 6.03 \times 10^{-3}T + 926{,}255T^{-2} - 968T^{-0.5} + 7663T^{-1}\right) dT \\
& + \int_{1644}^{1674} \left(4.31 + 1.28 \times 10^{-3}T + 320{,}782T^{-2} - 609T^{-0.5} + 7663T^{-1}\right) dT \\
& + \int_{1674}^{1700} \left(9.76 - 1.71 \times 10^{-3}T + 320{,}782T^{-2} - 609T^{-0.5} + 7663T^{-1}\right) dT \\
& + \int_{1700}^{1803} \left(-1.38 - 0.228 \times 10^{-3}T - 2{,}091{,}223T^{-2} - 154T^{-0.5} + 4993T^{-1}\right) dT \\
& + \int_{1803}^{T} \left(-1.68 - 0.228 \times 10^{-3}T - 2091{,}223T^{-2} - 154T^{-0.5} + 4993T^{-1}\right) dT \\
& + \Delta H_{Fe,\alpha-\beta} + \Delta H_{Fe,\beta-\gamma} + \Delta H_{Fe,\gamma-\delta} + \Delta H^{m}_{Fe,\delta-l} + \Delta H^{m}_{FeO} + \Delta H^{d}_{C}
\end{aligned}
$$

After integration:

$$
\begin{aligned}
\Delta H_R^o = {} & \Delta H_{R,298}^o \\
& + \left(14.74T + 0.105 \times 10^{-3}T^2 - 82{,}788T^{-1}\right. \\
& \quad \left. -1465T^{0.5} + 7663 \ln T - 789{,}289T^{-2}\right)\Big|_{298}^{1000} \\
& + \left(23.44T - 9.67 \times 10^{-3}T^2 - 969{,}255T^{-1} - 1936T^{0.5} + 7663 \ln T\right)\Big|_{1000}^{1033} \\
& + \left(30.47T - 4.51 \times 10^{-3}T^2 - 926{,}255T^{-1} - 1936T^{-0.5} + 7663 \ln T\right)\Big|_{1033}^{1179} \\
& + \left(24.92T - 3.01 \times 10^{-3}T^2 - 926{,}255T^{-1} - 1936T^{0.5} + 7663 \ln T\right)\Big|_{1179}^{1644} \\
& + \left(4.31T + 0.64 \times 10^{-3}T^2 - 320{,}782T^{-1} - 1218T^{0.5} + 7663 \ln T\right)\Big|_{1644}^{1674} \\
& + \left(9.76T - 0.85 \times 10^{-3}T^2 - 320{,}782T^{-1} - 1218T^{-0.5} + 7663 \ln T\right)\Big|_{1674}^{1700} \\
& + \left(-1.38T - 0.114 \times 10^{-3}T^2 + 2{,}091{,}223T^{-1} - 309T^{0.5} + 4993 \ln T\right)\Big|_{1700}^{1803} \\
& + \left(-1.68T - 0.114 \times 10^{-3}T^2 + 2{,}091{,}223T^{-2} - 309T^{-0.5} + 4993 \ln T\right)\Big|_{1803}^{T} \\
& + \Delta H_{Fe,\alpha-\beta} + \Delta H_{Fe,\beta-\gamma} + \Delta H_{Fe,\gamma-\delta} + \Delta H_{Fe,\delta-l}^m + \Delta H_{FeO}^m + \Delta H_C^d
\end{aligned}
$$

In the previous analysis C is in the form of C_{gr}. After considering the heat of dissolution: $\underline{C} \rightarrow C_{gr}$, and defining T as 1600 °C:

$$
\begin{aligned}
\Delta H_R^o &= 37{,}118.843 - 1118.17 - 13.4463 - 137.538 \\
&\quad - 1623.183 - 113.454 - 86.132 \\
&\quad - 341.501 - 253.31 + 410 + 210 \\
&\quad + 110 + 3700 + 7490 - 5100 \\
\Delta H_R^o &= 40{,}252.107 \ \tfrac{\text{cal}}{\text{mol}} \\
\Delta H_{R,C} &= \left(\tfrac{1000}{71.8464}\right)\Delta H_{R,ToK}^o = 560{,}252.246 \ \tfrac{\text{cal}}{\text{kg FeO}} \\
&= 559 \ \text{kcal/kg FeO}
\end{aligned}
$$

(2) $Fe_3C_{DRI} \rightarrow 3Fe + \underline{C}$

$$\Delta H_R^o = \Delta H_{R,\,298}^o + \int_{298}^{463} \Delta C_p dT + \int_{463}^{1000} \Delta C_p dT + \int_{1000}^{1033} \Delta C_p dT + \int_{1033}^{1179} \Delta C_p dT + \int_{1179}^{1500} \Delta C_p dT + \int_{1500}^{1674} \Delta C_p dT + \int_{1674}^{1803} \Delta C_p dT + \int_{1803}^{T} \Delta C_p dT + \Delta H_{Fe,\alpha-\beta} + \Delta H_{Fe,\beta-\gamma} + \Delta H_{Fe,\gamma-\delta}^s + \Delta H_{Fe,\delta-l}^m + \Delta H_{Fe_3C,\alpha-\beta} + \Delta H_{Fe_3C,\beta-l}^m + \Delta H_C^d$$

where:

$$\Delta C_p = 3C_{p,Fe} + C_{p,C} - C_{p,Fe_3C}$$

Replacing the heat capacities for each temperature range:

$$\begin{aligned}\Delta H_R^o = {} & \Delta H_{R,\,298}^o \\ & + \int_{298}^{463} \left(5.09 - 0.61 \times 10^{-3}T + 326{,}315T^{-2} - 244T^{-0.5} - 1{,}578{,}578T^{-3}\right) dT \\ & + \int_{463}^{1000} \left(-0.88 + 1.638 \times 10^{-3}T + 326{,}315T^{-2} - 244T^{-0.5} - 1{,}578{,}578T^{-3}\right) dT \\ & + \int_{1000}^{1033} \left(-9.58 + 18.52 \times 10^{-3}T - 560{,}151T^{-2} - 37.94T^{-0.5}\right) dT \\ & + \int_{1033}^{1179} \left(11.5 - 2.77 \times 10^{-3}T - 689{,}151T^{-2} - 37.94T^{-0.5}\right) dT \\ & + \int_{1179}^{1500} \left(-5.14 + 6.22 \times 10^{-3}T - 689{,}151T^{-2} - 37.94T^{-0.5}\right) dT \\ & + \int_{1500}^{1674} \left(-8.52 + 9.22 \times 10^{-3}T - 689{,}151T^{-2} - 37.94T^{-0.5}\right) dT \\ & + \int_{1674}^{1803} \left(7.82 + 0.228 \times 10^{-3}T - 689{,}151T^{-2} - 37.94T^{-0.5}\right) dT\end{aligned}$$

$$+\int_{1803}^{T}\left(6.92+0.228\times10^{-3}T-689{,}151T^{-2}-37.94T^{-0.5}\right)dT$$
$$+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H^{s}_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}$$
$$+\Delta H_{Fe_3C,\alpha-\beta}+\Delta H^{m}_{Fe_3C,\beta-l}+\Delta H^{d}_{C}$$

After integration:

$$\begin{aligned}\Delta H^{o}_{R}=\Delta H^{o}_{R,298}&\\
&+\left(5.09T-0.308\times10^{-3}T^{2}-326315T^{-1}\right.\\
&\quad\left.-489T^{0.5}+789{,}289T^{-2}\right)\Big|_{298}^{463}\\
&+\left(-0.88T+0.819\times10^{-3}T^{2}-326{,}315T^{-1}-489T^{0.5}+789{,}289T^{-2}\right)\Big|_{463}^{1000}\\
&+\left(-9.58T+9.26\times10^{-3}T^{2}+560{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1000}^{1033}\\
&+\left(11.5T-1.38\times10^{-3}T^{2}+689{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1033}^{1179}\\
&+\left(-5.14T+3.11\times10^{-3}T^{2}+689{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1179}^{1500}\\
&+\left(-8.52T+4.61\times10^{-3}T^{2}+689{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1500}^{1674}\\
&+\left(7.82T+0.114\times10^{-3}T^{2}+689{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1674}^{1803}\\
&+\left(6.92T+0.114\times10^{-3}T^{2}+689{,}151T^{-1}-18.97T^{0.5}\right)\Big|_{1803}^{T}\\
&+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H^{s}_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}\\
&+\Delta H_{Fe_3C,\alpha-\beta}+\Delta H^{m}_{Fe_3C,\beta-l}+\Delta H^{d}_{C}\end{aligned}$$

In the previous analysis C is in the form of C_{gr}. After considering the heat of dissolution: $C_{gr}\rightarrow\underline{C}$, and defining T as 1600 °C:

$$\begin{aligned}\Delta H^{o}_{R}&=-5400-896.121+1388.118\\
&\quad+6934.418+1107.595+817.484\\
&\quad+975.455+1001.629+484.253+410\\
&\quad+210+110+3700+180+12300+5100\\
\Delta H^{o}_{R}&=28,422.83\ \tfrac{cal}{mol}\\
\Delta H_{R,C,T^{\circ}K}&=\left(\tfrac{1000}{179.552}\right)\Delta H^{o}_{R,T^{\circ}K}=158{,}298.598\ \tfrac{cal}{kg\,Fe_3C}\end{aligned}$$

(3) $2FeO+\underline{Si}\rightarrow2Fe+SiO_2$

$$\Delta H^{o}_{R}=\Delta H^{o}_{R,298}+\int_{298}^{1033}\Delta C_p dT+\int_{1033}^{1179}\Delta C_p dT$$

$$+\int_{1179}^{1644}\Delta C_p dT+\int_{1644}^{1674}\Delta C_p dT+\int_{1674}^{1685}\Delta C_p dT$$
$$+\int_{1685}^{1803}\Delta C_p dT+\int_{1803}^{T}\Delta C_p dT$$
$$+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H_{Fe,\gamma-\delta}$$
$$+\Delta H^{m}_{Fe,\delta-l}+\Delta H^{m}_{FeO}+\Delta H^{m}_{Si}+\Delta H^{d}_{Si}$$

where:

$$\Delta C_p = 2C_{p,Fe}+C_{p,SiO_2}-2C_{p,FeO}-C_{p,Si}$$

Replacing the heat capacities for each temperature range:

$$\Delta H^{o}_{R}=\Delta H^{o}_{R,\,298}$$
$$+\int_{298}^{1033}\left(29.86-1.35\times10^{-3}T+794{,}468T^{-2}-807T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1033}^{1179}\left(43.92-15.54\times10^{-3}T+708{,}468T^{-2}-807T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1179}^{1644}\left(32.82-9.556\times10^{-3}T+708{,}468T^{-2}-807T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1644}^{1674}\left(-8.39+5.074\times10^{-3}T-502{,}478T^{-2}-89.55T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1674}^{1685}\left(2.508+0.925\times10^{-3}T-502{,}478T^{-2}-89.55T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1685}^{1803}\left(2.667+1.007\times10^{-3}T-586{,}845T^{-2}-89.55T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\int_{1803}^{T}\left(2.077+1.007\times10^{-3}T-586{,}845T^{-2}-89.55T^{-0.5}+66{,}938{,}861T^{-3}\right)dT$$
$$+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}+\Delta H^{m}_{FeO}+\Delta H^{m}_{Si}+\Delta H^{d}_{Si}$$

After integration:

$$
\begin{aligned}
\Delta H_R^o = {} & \Delta H_{R,\,298}^o \\
& + \left(29.8T - 0.678 \times 10^{-3}T^2 - 794{,}468T^{-1} - 1614T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{298}^{1033} \\
& + \left(-43.92T - 7.778 \times 10^{-3}T^2 - 708{,}468T^{-1} - 1614T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1033}^{1179} \\
& + \left(32.82T - 4.778 \times 10^{-3}T^2 - 708{,}468T^{-1} - 1614T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1179}^{1644} \\
& + \left(-8.39T + 2.537 \times 10^{-3}T^2 + 502{,}477T^{-1} - 179.107T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1644}^{1674} \\
& + \left(2.508T - 0.462 \times 10^{-3}T^2 + 502{,}477T^{-1} - 179.107T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1674}^{1685} \\
& + \left(2.667T - 0.503 \times 10^{-3}T^2 - 586{,}845T^{-1} - 179.107T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1685}^{1803} \\
& + \left(2.077T - 0.503 \times 10^{-3}T^2 - 586{,}845T^{-1} - 179.107T^{0.5} - 33{,}469{,}430T^{-2}\right)\Big|_{1803}^{T} \\
& + \Delta H_{Fe,\alpha-\beta} + \Delta H_{Fe,\beta-\gamma} + \Delta H_{Fe,\gamma-\delta} + \Delta H_{Fe,\delta-l}^m + \Delta H_{FeO}^m + \Delta H_{Si}^m + \Delta H_{Si}^d
\end{aligned}
$$

After considering the heat of dissolution of Si in liquid steel: $Si_l \rightarrow \underline{Si.}$ and defining T as 1600 °C:

$$
\begin{aligned}
\Delta H_R^o &= -89558.497 - 1277.104 + 410 + 210 \\
&\quad + 110 + 3700 + 7490 + 11{,}998 + 31{,}100 \\
\Delta H_R^o &= -35{,}817.513 \ \tfrac{cal}{mol} \\
\Delta H_{R,SiO_2} &= \left(\tfrac{1000}{28.0855}\right)\Delta H_{R,T^\circ K}^\circ = -1{,}275{,}302.69 \ \tfrac{cal}{kg\,Si}
\end{aligned}
$$

(4) $\underline{Si} + O_2 \rightarrow SiO_2$

$$
\Delta H_R^o = \Delta H_{R,\,298}^o + \int_{298}^{1000} \Delta C_p dT + \int_{1000}^{1685} \Delta C_p dT + \int_{1685}^{T} \Delta C_p dT + \Delta H_{Si}^m + \Delta H_{Si}^d
$$

where:

$$
\Delta C_p = C_{p,SiO_2} - C_{p,Si} - C_{p,O_2}
$$

Replacing the heat capacities for each temperature range:

$$\Delta H_R^o = \Delta H_{R,\,298}^o + \int_{298}^{1000} \left(8.07 - 4.98 \times 10^{-3}T - 557{,}288T^{-2} - 70.63T^{-0.5} + 66{,}938{,}861T^{-3} + 16.17 \times 10^{-7}T^2\right) dT + \int_{1000}^{1685} \left(-6.92 - 0.578 \times 10^{-3}T + 3{,}962{,}791T^{-2} + 896.71T^{-0.5} + 66{,}938{,}861T^{-3} - 22{,}897T^{-1}\right) dT + \int_{1685}^{T} \left(-7.97 + 0.345 \times 10^{-3}T + 3{,}878{,}423T^{-2} + 896.71T^{-0.5} + 66{,}938{,}861T^{-3} - 22{,}897T^{-1}\right) dT + \Delta H_{Si}^m + \Delta H_{Si}^d$$

After integration:

$$\Delta H_R^o = \Delta H_{R,\,298}^o + \left(8.07T - 2.49 \times 10^{-3}T^2 + 557{,}288T^{-1} - 141.26T^{0.5} - 33{,}469{,}430T^{-2} + 5.39 \times 10^{-7}T^3\right)\Big|_{298}^{1000} + \left(-6.92T - 0.289 \times 10^{-3}T^2 - 3{,}962{,}791T^{-1} + 1793.17T^{0.5} - 33{,}469{,}430T^{-2} - 22{,}897 \ln T\right)\Big|_{1000}^{1685} + \left(-7.97T + 0.172 \times 10^{-3}T^2 - 3{,}878{,}423T^{-1} + 1793.17T^{0.5} - 33{,}469{,}430T^{-2} - 22{,}897 \ln T\right)\Big|_{1685}^{T} + \Delta H_{Si}^m + \Delta H_{Si}^d$$

After considering the heat of dissolution of Si in liquid steel: $Si_l \rightarrow \underline{Si}$ and defining T as 1600 °C:

$$\Delta H_R^o = -216{,}619.357 + 924.222 + 1313.533 + 426.119 + 11{,}998 + 31{,}100$$

$$\Delta H_R^o = -170{,}857.483 \frac{cal}{mol}$$

$$\Delta H_{R,\,SiO_2} = \left(\frac{1000}{60.0843}\right)\Delta H_{R,T^\circ K}^o = -2{,}843{,}629.417 \frac{cal}{kg\ SiO_2}$$

(5) $\underline{C} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$

$$\Delta H_R^o = \Delta H_{R,\,298}^o + \int_{298}^{1000} \Delta C_p dT + \int_{1000}^{1700} \Delta C_p dT + \int_{1700}^{T} \Delta C_p dT + \Delta H_C^d$$

where:

$$\Delta C_p = C_{p,CO} - C_{p,C} - \frac{1}{2}C_{p,O_2}$$

Replacing the heat capacities for each temperature range:

$$\begin{aligned}\Delta H_R^0 &= \Delta H_{R,298}^0 \\ &+ \int_{298}^{1000} \left(3.85 - 1.6\times10^{-3}T - 593{,}090T^{-2} - 364.6T^{-0.5} + 1{,}578{,}579T^{-3} + 8\times10^{-7}T^2 + 7663T^{-1}\right)dT \\ &+ \int_{1000}^{1700} \left(5.05 - 1.54\times10^{-3}T + 2{,}553{,}416T^{-2} - 116.22T^{-0.5} - 3785T^{-1}\right)dT \\ &+ \int_{1700}^{T} \left(-6.102 - 0.056\times10^{-3}T + 141{,}411T^{-2} + 338.35T^{-0.5} - 6455T^{-1}\right)dT + \Delta H_C^d\end{aligned}$$

After integration:

$$\begin{aligned}\Delta H_R^o &= \Delta H_{R,298}^o \\ &+ \left(3.85T - 0.8\times10^{-3}T^2 + 593090T^{-1} - 729T^{0.5}\right. \\ &\quad \left. -789289T^{-2} + 2.7\times10^{-7}T^3 + 7663\ln T\right)\Big|_{298}^{1000} \\ &+ \left(5.05T - 0.773\times10^{-3}T^2 - 2553416T^{-1} - 232.45T^{0.5} - 3785\ln T\right)\Big|_{1000}^{1700}\end{aligned}$$

In the previous analysis C is in the form of C_{gr}: $C_{gr} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$. After considering this reaction: $\underline{C} \rightarrow C_{gr}$ and defining final T as 1600 °C:

$$\begin{aligned}\Delta H_{R,\,1873} &= -26{,}416.587 - \cdots - 5100 \\ \Delta H_R^o &= -26{,}416.587 - 348.472 - 1116.191 - 305.663 - 5100 \\ \Delta H_R^o &= -33{,}286.91\,\tfrac{cal}{mol} \\ \Delta H_{R,\,1873} &= \left(\frac{1000}{28.01}\right)\Delta H_{R,T}^o = -1{,}188{,}393.79\,\frac{cal}{kg\,CO}\end{aligned}$$

(6) $Fe + \frac{1}{2}O_2 \rightarrow FeO$

$$\begin{aligned}\Delta H_R^o = \Delta H_{R,\,298}^o &+ \int_{298}^{1000} \Delta C_p\,dT + \int_{1000}^{1033} \Delta C_p\,dT \\ &+ \int_{1033}^{1179} \Delta C_p\,dT + \int_{1179}^{1644} \Delta C_p\,dT + \int_{1644}^{1674} \Delta C_p\,dT \\ &+ \int_{1674}^{1700} \Delta C_p\,dT + \int_{1700}^{1803} \Delta C_p\,dT\end{aligned}$$

$$+\int_{1803}^{T}\Delta C_p\,dT+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}$$
$$+\Delta H_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}+\Delta H^{m}_{FeO}$$

where:

$$\Delta C_p=C_{p,FeO}-C_{p,Fe}-\tfrac{1}{2}C_{p,O_2}$$

Replacing the heat capacities for each temperature range:

$$\begin{aligned}\Delta H^{o}_{R}&=\Delta H^{o}_{R,\,298}\\&+\int_{298}^{1000}\left(10.89-1.8\times10^{-3}T-589,878.7T^{-2}+368.18T^{-0.5}+8.085\times10^{-7}T^{2}\right)dT\\&+\int_{1000}^{1033}\left(-18.39+0.388\times10^{-3}T+1{,}670{,}161T^{-2}+851.855T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\int_{1033}^{1179}\left(-25.42+7.48\times10^{-3}T+1{,}670{,}161T^{-2}+851.855T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\int_{1179}^{1644}\left(-19.87+4.48\times10^{-3}T+1{,}670{,}161T^{-2}+851.855T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\int_{1644}^{1674}\left(0.732-2.82\times10^{-3}T+2{,}232{,}634T^{-2}+493.133T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\int_{1674}^{1803}\left(-4.717+0.173\times10^{-3}T+2{,}232{,}634T^{-2}+493.133T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\int_{1803}^{T}\left(-4.417+0.173\times10^{-3}T+2{,}232{,}634T^{-2}+493.133T^{-0.5}-11{,}448T^{-1}\right)dT\\&+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}+\Delta H^{m}_{FeO}\end{aligned}$$

After integration:

$$\begin{aligned}\Delta H^{o}_{R}&=\Delta H^{o}_{R,\,298}\\&+\left(10.89T-0.906\times10^{-3}T^{2}+589,878.7T^{-1}+736.36T^{0.5}+2.695\times10^{-7}T^{3}\right)\Big|_{298}^{1000}\\&+\left(-18.39T+0.194\times10^{-3}T^{2}-1{,}670{,}161T^{-1}+1703.71T^{-0.5}-11,448\ln T\right)\Big|_{1000}^{1033}\\&+\left(-25.42T-4.51\times10^{-3}T^{2}-1{,}670{,}161T^{-1}+1703.71T^{0.5}-11{,}448\ln T\right)\Big|_{1033}^{1179}\end{aligned}$$

$$+\left(-19.87T+2.24\times10^{-3}T^{2}-1{,}670{,}161T^{-1}+1703.71T^{0.5}-11{,}448\ln T\right)\Big|_{1179}^{1644}$$
$$+\left(0.732T-1.41\times10^{-3}T^{2}-2{,}232{,}634T^{-1}-986.26T^{0.5}-11{,}448\ln T\right)\Big|_{1644}^{1674}$$
$$+\left(-4.717T+0.086\times10^{-3}T^{2}-2{,}232{,}634T^{-1}-986.26T^{0.5}-11{,}448\ln T\right)\Big|_{1674}^{1803}$$
$$+\left(-4.417T+0.086\times10^{-3}T^{2}-2{,}232{,}634T^{-1}-986.26T^{0.5}-11{,}448\ln T\right)\Big|_{1803}^{T}$$
$$+\Delta H_{Fe,\alpha-\beta}+\Delta H_{Fe,\beta-\gamma}+\Delta H_{Fe,\gamma-\delta}+\Delta H^{m}_{Fe,\delta-l}+\Delta H^{m}_{FeO}$$

for T = 1600 °C:

$$\Delta H^{o}_{R}=-63{,}535.43+3972.287-30.624-79.546$$
$$+867.628+61.801+31.561$$
$$+128.46+410+210+110$$
$$+3700+7490$$
$$\Delta H^{o}_{R}=-49{,}663.96\,\tfrac{cal}{mol}$$
$$\Delta H_{R,FeO,T^{\circ}K}=\left(\tfrac{1000}{71.8464}\right)\Delta H^{\circ}_{R,T^{\circ}K}=-691{,}250.501\,\tfrac{cal}{kg\,FeO}$$

Table 2.14 summarizes the heat of reaction of the few selected chemical reactions.

Comparing values for the heat of reaction requires a clear definition of the physical state of reactant species and temperature, for example, for the C–O reaction in the previous table it is clearly indicated that carbon is dissolved in liquid steel and oxygen is in the form of a gas. The result is different if carbon is solid or oxygen is also dissolved in liquid steel. Table 2.15 indicates values reported by Pfeifer and Kirschen [5] and Jones [7] for the heat of oxidation reactions.

Table 2.14 Heats of reaction of selected chemical reactions

	Reaction	$\Delta H^{0}_{R,1873}$, cal/mol	$\Delta H^{0}_{R,1873}$, cal/kg oxide	$\Delta H^{0}_{R,1873}$, kWh/kg oxide	$\Delta H^{0}_{R,1873}$, kWh/kg element
ΔH_{R1}	$FeO_{DRI}+\underline{C}\rightarrow Fe+CO$	40,252.1	560,252.2	0.65	
ΔH_{R2}	$Fe_3C_{DRI}\rightarrow 3Fe+\underline{C}$	28,422	158,298.5	0.18	
ΔH_{R3}	$2FeO+\underline{Si}\rightarrow 2Fe+SiO_2$	− 35,817.5	− 1,275,302.6	− 1.48	
ΔH_{R4}	$\underline{Si}+O_2\rightarrow SiO_2$	− 170,857.4	− 2,843,629.4	− 3.30	− 7.07
ΔH_{R5}	$\underline{C}+\frac{1}{2}O_{2(g)}\rightarrow CO_{(g)}$	− 33,286.91	− 1,188,393.7	− 1.38	− 3.22
ΔH_{R6}	$Fe+\frac{1}{2}O_2\rightarrow FeO$	− 49,663.86	− 691,250.5	− 0.80	− 1.03

Table 2.15 Heats of oxidation reactions

Reaction	ΔH_R^0, kWh/kg element [5]	ΔH_R^0, kWh/m^3 O_2 [5]	ΔH_R^0, kWh/m^3 O_2 @1593 °C, [7]
$Si + O_2 = SiO_2$	– 8.70	– 10.92	– 9.182
$C + 0.5O_2 = CO$	– 2.55	– 2.73	– 3.178
$C + O_2 = CO_2$	– 9.10	– 4.88	
$Fe + 0.5O_2 = FeO$	– 1.32	– 6.58	– 6.003
$2Fe + 1.5O_2 = Fe_2O_3$	– 2.03	– 4.74	
$2Al + 1.5O_2 = Al_2O_3$	– 8.61	– 13.86	– 11.936
$Mn + 0.5O_2 = MnO$	– 1.95	– 9.56	– 8.404
$S + O_2 = SO_2$	– 2.75	– 3.94	

Chemical energy from burners can be represented by the following chemical reaction:

$$CH_{4(g)} + 2O_{2(g)} = CO_{2(g)} + 2H_2O_{(g)}$$
$$\Delta H_{1823\,K} = -404\,\text{kJ/mol} = 10\,\text{kWh/m}^3$$

Chemical energy from post-combustion can be represented by the following chemical reaction:

$$CO_{(g)} + {}^1/_2O_{2(g)} = CO_{2(g)}$$
$$\Delta H_{1823\,K} = -556\,\text{kJ/mol}$$

Sims [31] provided the following values for the heat of formation of oxides. The large differences cannot be attributed, in principle, to the thermodynamic databases but most probably to the physical states of the reactants which is not always clearly stated in most references.

Oxide	CaO	MgO	SiO_2	FeO	Fe_2O_3	MnO	Al_2O_3	P_2O_5	CO
ΔH_f^0, kWh/kg	– 3.148	– 4.149	– 4.206	– 1.028	– 1.429	– 1.509	– 4.567	– 3.007	– 1.097

Table 2.16 Heat of formation of dicalcium silicate

Phase	H, cal/mol	Range, K	R^2
$2CaO \cdot SiO_2$	$0.11907\,T - 30,225.50197$	298–970	1.000
	$3.433 \times 10^{-3}T^2 - 5.632322T - 27,406.500666$	970–1710	0.9983
	$4.049168T - 30,493.638427$	1710–2000	0.9995

E. G. King, *J. Am. Chem. Soc.*, 73, 656, 1951.
J. F. Elliot, M. Gleiser, V. Ramakrishna, *Thermochemistry for Steelmaking*, Vol. 2, 1963, pp. 301.

Table 2.17 Heat of formation of slag species, after Sims [31]

Phase	ΔH_f^0, kWh/kg	Source
$2CaO{\cdot}SiO_2$	-5.846×10^{-1}	SiO_2
$2FeO{\cdot}SiO_2$	-6.637×10^{-2}	FeO
$Fe_2O_3{\cdot}2CaO$	-5.389×10^{-2}	Fe_2O_3
$4CaO{\cdot}P_2O_5$	-1.315×10^{0}	P_2O_5
$3CaO{\cdot}Al_2O_3$	-4.220×10^{-2}	Al_2O_3

The slag is formed from several sources; oxides present in the gangue content of DRI, scrap dust, oxides formed due to oxidation of elements in scrap, etc. The slag is not composed by pure oxides but defined by a complex multi-component equilibrium. Table 2.16 summarizes some correlations to define the heat of formation of dicalcium silicate and Table 2.17 provides additional information from other oxide species.

The melting point of slag components is in general above the normal steelmaking temperature (1600 °C). Since the slag is in fact liquid below that temperature is an indication of the presence of a complex multicomponent system, dominated by the presence of FeO, CaO and SiO_2 which form lower melting point compounds. To simplify the calculations is it assumed an average chemical composition. Daiconu et al. estimated a melting enthalpy, ΔH_m^0, of 5.8×10^{-2} kWh/kg.

2.3.5 Case Studies with Static Energy Balances

Prakash et al. [2] reported mass and energy balances to study the effect of the fraction of DRI and hot metal in the metallic charge on energy consumption. In a heat balance the energy inlets are electric and chemical energy and the energy outlets are liquid steel, liquid slag, the off-gas, water cooling, heat losses through the walls, electric losses and other heat losses. Their results suggest an increment in electric energy of 10 kWh/ton for every 10% increment in DRI. The highest consumption of electric energy and lowest input of chemical energy, 83% and 17% respectively, was found for the case using 100% scrap. They suggested to increase the fraction of hot metal or an increase in the carbon content in DRI to decrease the consumption of electric energy.

Arzpeyama et al. [30] applied energy balances to study the effect of HBI on oxygen injection, energy consumption and slag volume. They estimated the metal and slag composition using distribution factors which depend on the equilibrium partition ratios. Their results indicate that increasing 1% HBI decreases the consumption of oxygen and increases both energy consumption and the amount of slag by 0.16 Nm^3/ton, 1.29 kWh/ton and 34 kg/ton, respectively.

Diaconu et al. [19] described a detailed analysis of heat losses for a small EAF of 7 ton using scrap. They reported a low thermal efficiency of about 44% due to large heat losses, in particular the heat losses due to water cooling, however their numbers are inconsistent.

Kirschen et al. [40] evaluated heat balances for 16 EAF's charged with different ratios of scrap and DRI; 11 EAF's with 100% scrap and 5 EAF's with 70–90% DRI. They defined efficiency factors for heat transfer coming from the electric and chemical energies. Heat losses for each term was computed by the difference $(1-\eta_i)(\Delta H_i)$:

$$Q_m + Q_s = \eta_{elec}\Delta H_{elec}\tau + \eta_{chem}\Delta H_{chem}\tau + \eta_{burner}\Delta H_{burner}\tau + \eta_{pc}\Delta H_{pc}\tau$$

where: Q_m is the sensible heat in liquid steel, Q_s is the sensible heat in liquid slag, τ is time.

The electric efficiency term included not only Joule losses but also radiation at the electric arcs ($\eta_{el}\eta_{arc}$).

By assigning the efficiencies indicated in Table 2.18 they were able to reproduce the experimental values. These results are very useful, however, it should also be noticed that each heat evaluated was assigned a different set of efficiencies. The efficiency factors change for every heat. For example, similar heats with 100% scrap reported an oxygen efficiency from 76 to 90%. In other words, this information is only a reference which cannot be employed to reproduce any heat. Also notice that the values on efficiencies for burners and post-combustion for heats employing high amounts of DRI is wrong because those methods only apply for scrap-based operations.

Considering the values reported in Table 2.18 for the combined electric efficiency, $\eta_{el}\eta_{arc}$, it can be estimated the arc efficiency. Pfeifer et al. [6] reported electric efficiency at the secondary side of the transformer in the range from 0.88 to 0.92. Heat transfer efficiency from the arc to the molten metal depends on the slag foaming conditions and can fluctuate from 0.36 to 0.93. According with the data in the table,

Table 2.18 Heat transfer efficiencies using scrap and DRI in the EAF [40]

	$\eta_{el}\eta_{arc}$	η_{chem}	η_{oxygen}	η_{burner}	$\eta_{post\text{-}comb}$
70–90% DRI	65–68	73–75	82–90	55	10
100% DRI	67–80	70–75	76–90	55	10
Range	65–80	70–75	76–90	55	10

Table 2.19 Total energy consumption and distribution of energy (%)

Refs.	Metallic charge	TE	EE	CE	Steel	Slag	OffG	ee	wc	rad
[2]	60DRI/40S	672	83	17	43.2	7.0	13.6	11.7	5.8	17.5
[2]	70HM/ 20DRI/10S	674	62	37	43.2	5.3	14.8	11.6	5.8	17.5
[2]	40HM/ 20DRI/20PI/ $20S^{real}$	767	45	54	38	6	16	12	7	19
[2]	40HM/ 20DRI/20PI/ $20S^{mod}$	674	47	57	43	5.3	14.8	11.6	5.8	17.5
[2]	Scrap	597	83	17	48.8	4.9	10.6	11.6	5.8	17.5
[5]	Scrap	758	62	38	53	5.4	16.9	3.8	18.9	
[5]	Scrap	810	48	52	53	3.8	20.7		13	
[19]	Scrap				44	8	6.7	3	22	
	Scrap	762	61.2	38.8	51.4	5.5	15.9	4.6	15	7.6
[40]	Scrap/DRI	650–850	40–65	22–60	45–60	5–10	15–35		10–20	

TE = total energy, EE = electric energy, CE = chemical energy, Steel = sensible heat in liquid Steel, Slag = sensible heat ion liquid slag, OffG = sensible heat in the off-gas, ee = heat losses electric circuit, wc = heat losses due to water cooling, rad = heat losses due to radiation

the combined efficiency is about 65–80%. Taking an average value for $\eta_{el} = 0.9$, the average value of $\eta_{arc} = 0.7$–0.9 for the heats evaluated.

Table 2.19 summarizes reported energy distributions using different raw materials and proportions of chemical energy from different sources. It can be observed that heat losses in the off-gas range from 10 to 20%, water cooling from 6 to 20% and radiation from 17 to 19%, this last term strongly influenced by the number of charges of scrap baskets. Chemical energy for normal scrap and DRI operation without burners and post-combustion is in the order of 17%, increasing up to 70% as hot metal, burners and post-combustion is included in the heat balance, for the group of data in this table.

2.4 Dynamic Energy Balance

A dynamic energy balance involves rate phenomena, defines the instantaneous changes in chemical composition of all phases (liquid steel, liquid slag, off-gas, dust), their temperature and their mass. It involves a much more complex analysis in comparison with a static model. In addition to thermodynamics the rates of heat, momentum and mass should be considered. A dynamic model describes the rates of change in all the variables and therefore the problem involves differential equations which can be solved using standard methods or computational fluid-dynamics (CFD)

software, in all cases involving the development of mathematical models. Realistic modelling of each single operation, for example; scrap melting, slag foaming, decarburization, etc. is usually impossible because of limitations on computing time or lack of information on kinetic parameters, and the only way to make a model is to simplify the number of variables and make assumptions. Even under these conditions, complex dynamic models are not adequate for control on-line due to the long time to run a single simulation. Model validation is also a challenge due to the difficulty to obtain direct information from the actual process, therefore validations are based on physical models or partial data taken from the real process.

Hay et al. [46] reviewed mathematical process models up to 2021. The majority of the models developed have focused on single operations. Several dynamic models based on proprietary software that include multiple operations have been reported [47–50], however the lack of details makes difficult to evaluate its real capabilities. In one of those cases, the full energy losses were not computed but using lump values which clearly is far from a phenomenological approach. Much more useful, in terms of free knowledge, are the reports from academic groups [51–55]. A summary of these works is described below which provides a better idea of the development of a dynamic model.

The work carried out at ESIQIE-IPN [51–53] in Mexico was one of the first research works reporting integral models that described in detail the reaction kinetics involving multiple operations using DRI in the EAF, in particular slag foaming. For full details the reader should check those references.

Figure 2.4 shows a typical EAF with continuous DRI feeding, oxygen and carbon injection. Their model describes the change in chemical composition of both slag and steel, the temperature of liquid steel and a novel dynamic foaming index. The DRI melting rate was described using an empirical relationship based on plant data.

Figure 2.5 shows schematically the general structure of the mathematical model using 100% DRI. Energy consumption is initially estimated from an energy balance but heat losses are largely under-estimated, because the off-gas composition is assumed to be formed only by CO and other heat-losses were also not considered.

The mass balance defines the mass of DRI, fluxes, coke and oxygen injection. Using the following empirical relationships for the DRI melting rate and lime dissolution rate it is possible to calculate the melting rate of DRI and rate of dissolution of fluxes:

$$\frac{dW_{DRI}}{dt} = -226{,}403.57 + 5016.6(\%Met) - 28.8(\%Met)^2$$

$$\frac{dW_{lime}}{dt} = f(T) \cdot X_{CaO}^{lime} \frac{dW_s}{dt}$$

where: W_{lime} is the rate of dissolution of lime, W_{DRI} is the mass of DRI, %Met is the degree of metallization of DRI, W_s is the mass of slag, f(T) is a function of temperature.

The rate of formation of CaO, SiO_2, MgO and Al_2O_3 depends on the rate of melting of DRI and the rate of dissolution of lime. Oxidation of elements is neglected because

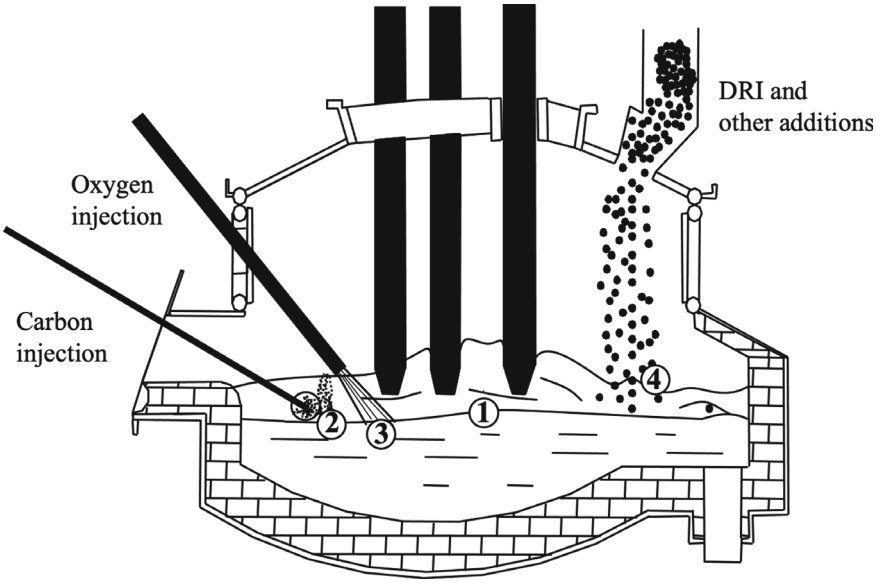


Fig. 2.4 EAF showing the reaction interfaces; (1) slag-metal interface, (2) carbon particles-slag interface, (3) oxygen injection-slag interface, (4) melting of additions

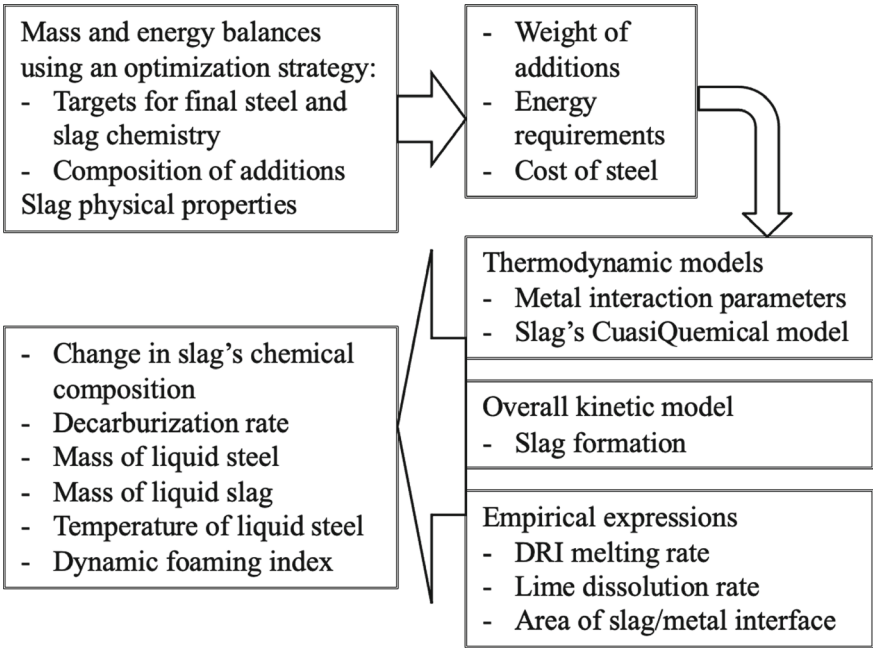


Fig. 2.5 Overview of dynamic model. After ref. [51–53]

there is no scrap in the metallic charge, only DRI.

$$\frac{d\left(X^{s}_{MO} \cdot W_s\right)}{dt} = X^{lime}_{MO}\frac{dW_{lime}}{dt} + X^{DRI}_{MO}\frac{dW_{DRI}}{dt}$$

where: X^{s}_{MO} is the fraction of oxide in the slag, X^{DRI}_{MO} is the fraction of the same oxide in DRI.

The final mass of iron oxide in the slag results from the FeO added by DRI and FeO due to oxidation minus the FeO reduced with carbon particles. The reduction rate of FeO by carbon particles is considered to take place by two simultaneous reactions; reduction at the CO/slag interface and consumption of carbon particles by the Boudouard reaction.

$$\frac{d\left(X^{s}_{FeO} \cdot W_s\right)}{dt} = X^{DRI}_{MO}\frac{dW_{DRI}}{dt} + \frac{M_{FeO}}{M_O}K_{O_2}W_{O_2} - M_{FeO}A_{sm}r''_{CO} - \frac{M_{FeO}}{M_O}Q_s\theta^{g}_{p}\left[1 - \left(\frac{r^{f}_{p}}{r^{0}_{p}}\right)^3\right]$$

where: M is the molecular mass, K_{O_2} is the rate of oxidation of liquid iron, r''_{CO} is the rate of formation of CO, Q_s is the air flow rate to inject carbon particles, θ^{g}_{p} is the fraction of particles that are able to penetrate into the slag, r is the radius of the carbon particles.

The rate of formation of slag is then given by the net formation of all oxides:

$$\frac{dW_s}{dt} = \frac{d\left(X^{s}_{FeO} \cdot W_s\right)}{dt} + \frac{d\left(X^{s}_{CaO} \cdot W_s\right)}{dt} + \frac{d\left(X^{s}_{SiO_2} \cdot W_s\right)}{dt} + \frac{d\left(X^{s}_{MgO} \cdot W_s\right)}{dt} + \frac{d\left(X^{s}_{Al_2O_3} \cdot W_s\right)}{dt}$$

The expression that defines the fraction of carbon particles that penetrate into the slag, θ^{g}_{p}, is critical to the calculation of both iron oxide reduction and the formation of CO. The expression developed was the following [51–53]:

$$\theta^{g}_{p} = 1 - \frac{12\sigma_{lg}(f - \cos\beta)}{\left(U^{g}_{p}\right)^2 d_p \rho_p}$$

where: f is the difference between the gas–liquid interface area while the particle is passing through the interface and the area of the undisturbed interface, σ_{lg} is the gas phase interfacial energy, β is the solid–liquid contact angle, d_p is the carbon particle diameter and ρ_p is the carbon particle density. The expression indicates that in order to maximize the fraction of particles crossing into the slag, the variables in the denominator of the second term on the right side of the equation should be

increased as much as possible and/or those in the numerator decreased. This result has significant consequences because with one single equation is possible to identify how to manipulate the process variables to improve the slag foaming conditions.

Another dynamic model reported by Bekker et al. [56] in 1999 was one of the first dynamic models that described the scrap melting rate and the composition of the off-gas. This work is based on thermodynamics and a large number of heat transfer approximations, giving a very rough representation of the real process. For example, the melting rate of solid scrap is estimated based on the temperature difference between solid scrap and liquid steel multiplied by a heat transfer factor proportional to the ratio of the same temperatures, in turn the temperature of liquid steel is estimated by dividing the total power in the furnace by the specific heat capacity of all materials heated to the melting temperature:

$$\frac{dW_{scrap}^{solid}}{dt} = -\frac{M_{Fe}k_t k_A W_{scrap}(T_l - T_s)\left(\sqrt{\frac{T_s}{T_l}}\right)}{\left(\lambda_{Fe} + C_{p,Fe}(T_l - T_s)\right)} = -\frac{dW_{steel}^{liquid}}{dt}$$

$$\frac{dT_l}{dt} = \frac{\sum \text{net power}}{\sum \frac{C_{pi}}{M_i}}$$

where: dW/dt is the rate of scrap melting, k_t is a heat transfer coefficient (0.4 kW/Km^2), k_A is the specific surface area of steel scrap (0.005 m^2/kg), T_l is the temperature of liquid steel and T_s is the temperature of scrap, λ_{Fe} is the latent heat of melting, C_p represents the heat capacity.

In their model, they assume the slag/metal system is far from equilibrium and the exact value of the equilibrium constants is not critical [56, 57]. This assumption eliminates the use of thermodynamic models to compute activity coefficients. They suggested an original approach to estimate the equilibrium composition. For example, for the following chemical reaction:

$$(FeO) + [C] = [Fe] + CO_g$$

The solubility product at equilibrium is:

$$k_{C\%} = (\%FeO)(\%C)_{eq}$$

$$k_{C\%} = \left(X_{FeO}\frac{M_{FeO}}{M_{slag}}100\right)\left(X_C^{eq}\frac{M_C}{M_{Fe}}100\right)$$

$$k_{C\%} = X_{FeO}X_C^{eq}\left(\frac{M_{FeO}M_C}{M_{slag}M_{Fe}}\right)100^2$$

From the previous expression, an expression for the equilibrium carbon concentration can be derived:

$$X_C^{eq} = \frac{k_{C\%}}{X_{FeO}}\left(\frac{M_{slag}M_{Fe}}{M_{FeO}M_C 100^2}\right)$$

The parameters which are constant are defined as a dimensionless equilibrium concentration:

$$k_{XC} = k_{C\%}\left(\frac{M_{slag}M_{Fe}}{M_{FeO}M_C 100^2}\right)$$
$$k_{XC} = X_{FeO} \times X_C^{eq}$$

In this way, these authors calculated the equilibrium carbon concentration as follows:

$$X_C^{eq} = \frac{k_{XC}}{X_{FeO}}$$

The value of $k_{C\%}$ was taken from Turkdogan [58]. The equilibrium constant given by Turkdogan can be calculated by the following expression:

$$\log K_{XC} = \log\left(\frac{p_{CO}}{[\%C]a_{FeO}}\right) = \frac{5730}{T} + 5.096$$

At 1600 °C, $K_{XC} = 108.8$. Turkdogan defined a_{FeO} in terms of (%FeO), assuming a slag with a binary basicity of 3.2, γ_{FeO}= 1.3. He reported the following calculations.

$$a_{FeO} = \gamma_{FeO}X_{FeO} = 1.3X_{FeO} = 1.3\frac{(\%FeO)}{72 \times 1.65} = 0.11 \times (\%FeO)$$
$$K_{XC} = \frac{p_{CO}}{[\%C]a_{FeO}} = \frac{1.5}{[\%C] \times 0.11 \times (\%FeO)}$$
$$[\%C] \times (\%FeO) = \frac{1.5}{0.11 \times K_{XC}} = \frac{1.5}{0.11 \times 108.8} = 1.25$$

This calculation has probably a typographical mistake because in both cases the correct math with those values is not 0.11 and 1.25. There is no attempt to correct it because these values have been employed subsequently in other models. With the previous result, is now possible to define the value k_{XC} in Bekker's model:

$$k_{C\%} = [\%C] \times (\%FeO) = 1.25$$
$$M_{FeO} = 0.0718\,kg/mol$$
$$M_C = 0.012\,kg/mol$$
$$M_{Fe} = 0.0558\,kg/mol$$
$$M_{slag} = 0.0606\,kg/mol$$
$$k_{XC} = k_{C\%}\left(\frac{M_{slag}M_{Fe}}{M_{FeO}M_C 100^2}\right)$$
$$= 1.25 \times \left(\frac{0.00338}{8.62}\right) = 490 \times 10^{-6}$$

$$k_{XC} = X_{FeO} \times X_C^{eq} = 490 \times 10^{-6}$$

This value is about one order of magnitude lower than the value reported by Turkdogan ($1/108 = 925 \times 10^{-5}$).

Bekker [57] and Oosthuizen et al. [59] extended the previous EAF model to study the off-gas composition and exit temperature. Bekker [57] originally assumed the off-gas is formed of $CO + CO_2 + N_2$. The net production rate of CO is the result of CO formed due to oxygen injection and reduction of iron oxide minus CO transformed to CO_2 due to post-combustion inside the EAF due to air ingress both at the slag door and the slip gap. It is assumed all the carbon particles are consumed.

$$\frac{dCO}{dt} = f\left[k_{DeC}\left(X_C - X_C^{eq}\right)\right] + f(\dot{m}_c) - f(k_{PR}) - f(D_{4th}h)$$

where: k_{DeC} is a decarburization rate constant (12 kg/s), X_C is the concentration of carbon in liquid steel, $\dot{m}_c$ is the rate of injection of carbon particles, k_{PR} is a constant which depends on the area of the slag door and the relative pressure inside the furnace, D_{4th} is the diameter of the fourth hole and the air-gap distance. The net production rate of CO_2 results from CO post-combustion inside the furnace and at the slip gap. Nitrogen depends on air ingress through the slag door and at the slip gap, and also air employed as a carrier gas for carbon injection. They estimated that for every 150 kg of injected graphite, there is 1 kg of nitrogen in the carrier gas.

In order to improve the accuracy of the model predictions on the exit temperature of the off-gas, Oosthuizen et al. [59] reviewed in more detail its heat balance considering the heat released due to combustion and the heat extracted due to convection through the duct sides. The amount of heat extracted was calculated using the Petukhov equation for turbulent convection heat transfer.

Matson and Ramirez [60] developed a dynamic model considering equilibrium at all the interfaces with the rates of reaction depending on transport limitations among the phases. The scrap melting rate was estimated assuming it was formed by spheres.

MacRosty and Swartz [54] developed a dynamic model aiming to achieve a phenomenological approach, capable to predict the nonlinear behavior of many variables, numerically robust and using the least amount of plant data. It consisted of 4 modules to study the melting rate of scrap, production of liquid steel, slag/metal reactions and the gas phase. All chemical reactions were assumed to be in equilibrium at the slag/metal interface, estimating the mass transfer coefficients with industrial data. The gas phase was composed by a complex mixture containing CO, CO_2, N_2, CH_4, H_2, H_2O and C_9H_{20}. The slag phase consists of FeO, MnO, MgO, Al_2O_3, SiO_2 and CaO. Due to the large number of equations in this model, the authors reported heat and mass balances at the interfaces in a general way. The first mass balance corresponds to the scrap zone where the rate of melting is computed:

$$\frac{dm_{ss}}{dt} = \dot{M}_{scrap} - \dot{M}_{melt}$$

where: m_{ss} is the mass of solid scrap, $\dot{M}_{scrap}$ is the mass of added scrap and $\dot{M}_{melt}$ is the melting rate of scrap. The melting rate was computed using a similar approach to that proposed by Bekker [57].

The mass balance at the molten metal zone, slag/metal interface, and gas zone are defined by the following general expression:

$$\frac{db_{k,z}}{dt} = F_{k,z}^{in} - F_{k,z}^{out}$$

where: b_k is the molar amount of element k in zone z, $F_{k,z}^{in}$ and $F_{k,z}^{out}$ are the flow rates of element k into and out of zone z.

An important suggestion from this work was the transformation into logarithmic space the molar quantities because the large differences in values would create numerical problems.

Heat balances were also reported with general expressions. The temperature of liquid steel was calculated similar to Bekker [57]. Due to the limited amount of experimental data (electrical and material inputs, off-gas chemistry, endpoint carbon and slag composition) the model predictions were compared with the off-gas, indicating a reasonably good accuracy.

The work from Logar et al. [25, 61] is one of the most updated dynamic models so far, extensively based on the previous work by Bekker [57] and MacRosty and Swartz (McR-S) [54] but described in much more detail and with additional improvements. It also describes the evolution in the chemical energy supplied over an entire heat. The slag is assumed as an ideal solution, formed by FeO, SiO_2, CaO, MnO, MgO, P_2O_5, Cr_2O_3 and Al_2O_3. Similar to McR-S, the geometric domain includes; the solid scrap zone, the liquid steel zone, the liquid slag zone, the gas zone, additionally the wall zone and roof zones were added. Every zone is assumed to be thermal and chemically homogeneous (perfect mixing conditions). This model does not compute the initial steel and slag chemistry for a given metallic charge but are provided as initial conditions.

The change in chemical composition for the metal, slag and gas was computed assuming that the driving force for a chemical reaction is the difference between the actual concentration and the equilibrium values, similar to Bekker:

$$\dot{m}_i = \frac{dm_i}{dt} = \frac{\Delta m_i}{\Delta t} = k_i\left(X_i - X_i^{eq}\right)$$

The net change in concentration varies for each chemical species depending on its sources and consumption rates. Carbon is added as carbon particles and other materials in the metallic charge, later on it can be consumed by FeO, MnO and oxygen injection. Silicon is a simpler case, it is consumed by oxidation with FeO, O_2 and MnO, as follows:

$$2(FeO) + [Si] = 2[Fe] + (SiO_2)$$
$$[Si] + O_{2(g)} = (SiO_2)$$

$$2(\mathrm{MnO}) + [\mathrm{Si}] = 2[\mathrm{Mn}] + (\mathrm{SiO_2})$$

$$\dot{m}_{Si} = \left[-k_{Si1}\left(X_{Si} - X_{Si}^{eq}\right)\right] + \left[-k_{Si2}\left(X_{Si} - X_{Si}^{eq}\right)\right]\left[k_{O_2-SiO_2}O_2\right] + \left[-k_{Si3}\left(X_{MnO} - X_{MnO}^{eq}\right)\frac{M_{Si}}{M_{MnO}}\right]$$

Once the mass of the different species for liquid metal, liquid slag and gas phase is defined as a function of time, then $\dot{m}_I$, the rate of chemical energy supplied and rate of energy consumed can be computed. Figure 2.6 shows the results reported by Logar et al. [54] for scrap melting. It shows the contribution to chemical energy by exothermic reactions (chemical), organic (combustible), oxidation of electrodes (electrodes) and burners (CH_4). According with these results, the main contribution is due to exothermic reactions of dissolved elements in the molten bath, about 26%. Oxidation of electrodes and burning of organic matter represent 2% and 1%, respectively. The chemical energy from oxy-fuel burner in this example was 9.2%.

The sensible heat in liquid steel, off-gas and slag was 51.4%, 15.9% and 5.5%, respectively. Heat losses by radiation was computed from the values of radiosity of N radiating surfaces, as follows:

$$Q_{i\text{-}rad} = A_i \sum_{j=1}^{N} VF_{ij}\left(J_i - J_j\right)$$

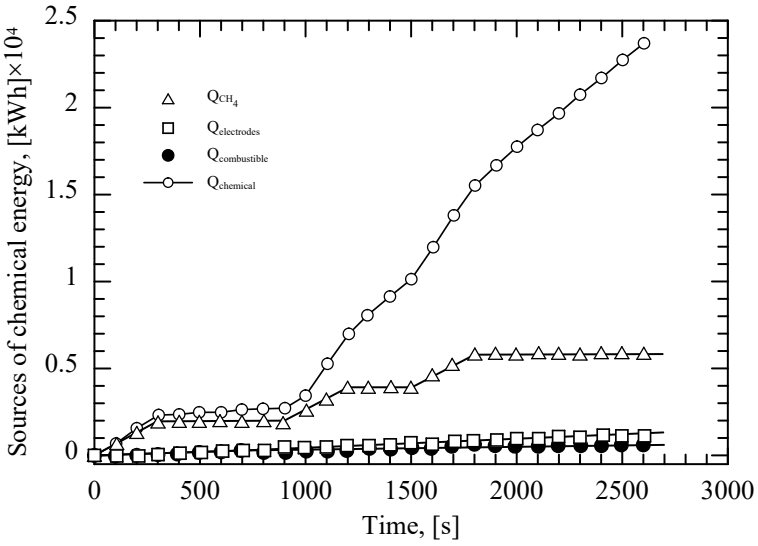


Fig. 2.6 Contribution of chemical energy throughout a heat. After Logar et al. [54]

$$J_i = \varepsilon_i \sigma_{SB} T_i^4 + (1 - \varepsilon_i) \sum_{j=1}^{N} \left(VF_{ij} - J_j \right)$$

where: J_i is the radiosity of body j, ε_i is the body's emissivity (0–1), σ_{SB} is the Stefan-Boltzmann constant, T_i is the body's temperature, VF_{ij} is the view factor from surface I to surface j (0–1). Heat transfer due to radiation to the water-cooled panels was estimated as 15%.

Other heat losses reported were; electric losses 4.6% and miscellaneous losses 7.6%.

According with Logar et al. [25, 61], their model accurately describes the evolution in chemical composition of steel, slag and the off-gas, however it should be noticed that agreement was largely influenced by the empirical parameters included in the model, valid for the specific conditions where the model was applied. Most of the empirical parameters are related to the mass transfer coefficients.

In addition to the extensive use of empirical parameters, it is quite clear that previous dynamic models have employed unrealistic assumptions like an ideal slag solution and perfect mixing conditions in the EAF which gives a tremendous room for further improvements. A research group at Aachen university [62] has incorporated the use of two slag models to predict thermodynamic activities of the slag components but their predictions show considerable deviations from the actual experimental values.

The Centre de recherches métallurgiques (CRM) in Belgium has reported the development of dynamic models for scrap melting [47, 63] and more recently have also included DRI in the metallic charge [64, 65]. Nyssen et al. [47, 63] provided general details about the dynamic model for scrap-based operations.

1. Scrap melting model: EAF is divided into 15 sectors, each sector with different layers
2. Bath model: Includes oxygen injection, oxidation reactions and follows the chemical, thermal and mass evolution of liquid steel. The model assumes perfect mixing conditions.
3. Slag model: Follows the chemical, thermal and mass evolution of liquid slag. The real slag foaming height is adjusted based on the slag leaving the furnace.
4. Refractory model: Calculates heat losses and erosion rate. The erosion rate is employed to compute the increment in volume available for slag and liquid steel.
5. Arc model: Defines heat transfer to different part of the EAF during scrap melting.
6. Scrap pre-heating: Computes heat exchange between off-gas and solid scrap.
7. Post-combustion: Calculates post-combustion including air ingress
8. Burners' module: Calculates the combustion of natural gas
9. Water panels module: Computes heat transfer to water cooled panels.
10. Skulls model: Evaluates solidification of hot heel due to long delays.

The exhaust flow rate is estimated with data at the quenching tower, using the amount of sprayed water and gas temperature before and after quenching, according with the following heat balance:

$$Q_{\text{off-gas}} C_p^{\text{off-gas}} \left(T^{f}_{\text{off - gas}} - T^{0}_{\text{off - gas}} \right) = Q_{\text{water}} \Delta H^{\text{vaporization}}_{\text{water}}$$

where: $Q_{\text{off-gas}}$ is the off-gas flow rate, Q_{water} is the flow rate of sprayed water.

The slag model predicts the foaming conditions. The slag volume is controlled with data from the refractory model that predicts the increment in volume due to erosion and the slag flow rate leaving the furnace. The furnace volume increases from heat to heat, for example considering an initial furnace capacity of 100 ton, after 280 heats they reported a furnace volume higher than 200 ton. It was reported that the slag volume remains almost constant during the heat. The apparent slag density of the foaming slag was calculated with the following equation:

$$\rho_s^{\text{app}} = \rho_s e^{-\frac{V_{CO}}{V_{CO}^{\text{ref}}}}$$

where: ρ_s^{app} is the apparent slag density, ρ_s is the slag density as a function of slag chemical composition, V_{CO} is the volume generated by the reduction of iron oxide V_{CO}^{ref} is a reference volume of CO.

The dynamic model developed by CRM using DRI assumes that DRI is heated instantaneously to 950 °C and FeO in DRI is self-reduced about 92% [65]. The models developed have been applied online for process control which might suggest the use of a large number of empirical factors obtained by trial and error at each plant.

2.5 Optimization Model Involving an Energy Balance

Another group of models is the use of optimization tools. The optimization schemes are usually focused on minimization or maximization of variables. There are two groups of optimization, single-objective optimization (SOO) and multiple-objective optimization (MOO). The simplest case is SOO, for example minimization of the price of steel, minimization of emissions of CO_2, minimization of energy consumption, etc. These types of problems have a single solution and easier to understand. More frequently the problem to be solved involves multiple-objective optimization (MOO) of several variables, for example minimization of the price of steel, minimization of energy consumption and minimization of emissions of CO_2. Deb [66] illustrates the following example which describes the differences between SOO and MOO: How to buy a car based on price or based on comfort, or both. If the decision is based on the minimum price, then the selection will be on the car with the lowest comfort, and vice versa. The two-objective optimization problem involves buying a car in which there is a trade-off between price and comfort. The trade-off involves

sacrificing objectives. Trade-off solutions are optimal solutions to a MOO problem, forming a front on an objective space, also called Pareto-optimal front.

Chapter 3 includes a summary of several optimization models. The following section describe in more detail the optimization models, in particular the one reported by Cardenas et al. [15].

2.5.1 Single-Objective Optimization (SOO)

The typical SOO in the steel industry has focused on the minimization of the production cost of steel. Once the objective function is defined the rest of the variables should align to reach this objective. Producing steel with the lowest production cost usually would affect other critical variables such as minimization of emissions of CO_2 and steel quality.

One of the first optimization models using linear programming was reported by Geiger in 1980 [67] which employed mixtures of scrap and DRI in the metallic charge. Cárdenas et al. [15] updated Geiger's model, and applied the model to study energy consumption with 100% DRI, applying the SIMPLEX method. Some of the improvements include the formation of CO_2, the addition of metallurgical and dolomitic lime, removing the addition of ferroalloys in the EAF and including DRI temperature. The system of equations is formed by the objective function, mass balances for the components in steel, slag and the off-gas, restrictions which consider special conditions such as availability of raw materials, maximum concentration of residual elements, etc., and a heat balance equation. Mass and heat balances are equalities and the restrictions are inequalities. The objective function is defined by the sum of costs for all materials and electric energy, as follows:

$$\text{Min } z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

Mass balances and restrictions:

$$\begin{aligned}
&A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n ?\, b_1\\
&A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n ?\, b_2\\
&\vdots \quad \ldots \quad \vdots\\
&A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n ?\, b_m
\end{aligned}$$

where: z is the cost of steel, c_i represents the cost of all materials and electric energy supplied, A_{ii} and A_{ij} are coefficients related to the fraction of a given chemical specie in the raw materials for the mass balances or specific energy consumption to raise the temperature of different raw materials as well as chemical energy, ? represents an operator $\leq$, $=$ or $\geq$.

The following example consists of 6 mass balances, 6 constraints and one energy balance. Mass balances include Fe, O, Si, Al, CaO, MgO, CO_2 and a general mass

balance. Si–Al and CaO–MgO are combined into one equation in each case. The mass balances for carbon and sulphur were handled as restrictions. Carbon is allowed to range between a minimum value to reach the carbon specifications in liquid steel and allow for the reduction of iron oxide in DRI to a maximum value which in addition to the previous two terms also include an amount of carbon for decarburization. The energy balance states input of electric energy (considering an electric efficiency of 83%) and the energy released due to exothermic reactions is equal to the heat required to raise the temperature of all materials to the steelmaking temperature plus sensible heat in the off gas, sensible heat in the slag, endothermic reactions and heat losses. Heat losses are considered in a lump parameter which depends on an empirical thermal efficiency. All equations are summarized in Table 2.20.

The commercial optimization software LINDO API® was employed. This software package solves linear programming, integer programming, nonlinear programming, stochastic programming and global optimization. The mass and energy balances were developed in a computer program using Visual Basic version 6.0 (currently obsolete) which has an interface to connect with the optimization software. The thermochemical data base was taken from the commercial software package FactSage®. The steelmaking temperature is defined as an inlet condition. The total number of unknowns is 10, including the weight of DRI and types of scrap, fluxes, oxygen and carbon requirements, weight of slag, weight of off-gas and consumption of electric energy. The solution defines the metallic charge that ensures minimum production cost for one ton of liquid steel. From a perspective of energy consumption, the model can evaluate the effect of chemical composition of raw materials, use of chemical energy, tapping temperature, scrap temperature and DRI temperature. Sensitivity analysis was carried out. A detailed explanation of the model is indicated in the thesis by Cárdenas et al. [44].

The model developed consists in three Graphical User Interfaces (GUI). The first one inputs the inlet conditions; chemical composition of all materials added to the EAF (DRI, scrap, fluxes and coke), the target chemistry of liquid steel, liquid slag and dust, tapping temperature, initial DRI temperature, maximum percentage of scrap and DRI (restrictions), cost of both, materials and electric energy in USD/kg and the thermal efficiency. The first interface computes all the coefficients in the system of equations, displayed on a second interface. This interface relates to the calculations using the SIMPLEX method and the results are shown on a third interface. The results provide the minimum cost per ton of steel, the amount of materials needed to produce one ton of steel and its electric energy consumption, it shows the mass balance, heat due to exothermic reactions, sensible heat in both liquid steel, liquid slag and the off-gas, and heat losses. The model can easily be adjusted to reach a higher accuracy in the thermal balance by changing the value of the thermal efficiency.

Figure 2.7 shows the results of a simulation describing the influence of the DRI ratio in the metallic charge on energy consumption, as a function of metallization of DRI and thermal efficiency. Increasing DRI in the metallic charge has a strong effect on energy consumption, however the intensity in this effect can be drastically reduced by increasing either the metallization of DRI or the thermal efficiency. The values shown in this figure indicate an increment of about 150 kWh/ton when the

Table 2.20 Set of equations for the optimization model

Impurity restriction: all impurities, including residual elements. This example; 0.4%
$X_{DRI}^{Imp} W_{DRI} + X_{Scrap}^{Imp} W_{Scrap} \leq 0.4\%$
Carbon restriction: Maximum C needed for DeC, as alloying element and to reduce FeO
$X_{DRI}^{C} W_{DRI} + X_{Scrap}^{C} W_{Scrap} + X_{Coke}^{C} W_{Coke} \leq 30 W_{Slag} + X_{Steel}^{C} W_{Steel} + X_{DRI}^{FeO} \left(\frac{12}{71.85}\right) W_{DRI}$
Scrap restriction: In the case that a given scrap needs to be below certain limit
$W_{Scrap} \leq Max$
Sulphur restriction: Maximum limit
$X_{DRI}^{S} W_{DRI} + X_{Scrap}^{S} W_{Scrap} + X_{Coke}^{S} W_{Coke} \leq X_{Steel}^{S} W_{Steel} + X_{Slag}^{S} W_{Slag}$
DRI restriction: Controls the fraction of DRI, from 0 to 100%
$W_{DRI} \leq Max$
Fe balance
$\left[X_{DRI}^{Fe} + \left(\frac{55.85}{71.85}\right) X_{DRI}^{FeO}\right] W_{DRI} + X_{Scrap}^{Fe} W_{Scrap} = \left[X_{Steel}^{Fe} + \left(\frac{55.85}{71.85}\right) X_{Steel}^{FeO}\right] W_{Steel} + \left(\frac{55.85}{71.85}\right) X_{Slag}^{FeO} W_{Slag} + \left(\frac{55.85}{71.85}\right) X_{Fume}^{FeO} W_{Fume}$
Si + Al balance: Si and Al in the charge are oxidized
$\left[\left(\frac{28.08}{60.08}\right) X_{DRI}^{SiO_2} + \left(\frac{53.96}{101.96}\right) X_{DRI}^{Al_2O_3}\right] W_{DRI} + X_{Scrap}^{Si+Al} W_{Scrap} = X_{Slag}^{Si+Al} W_{Slag}$
CaO + MgO Balance: CaO and MgO charged increase the volume of slag
$X_{DRI}^{CaO+MgO} W_{DRI} + X_{Cal}^{CaO+MgO} W_{Cal} = X_{Slag}^{CaO+MgO} W_{Slag}$
Oxygen balance: Oxygen in DRI and injected leaves in the gas phase and slag phase
$\left[\left(\frac{16}{71.85}\right) X_{DRI}^{FeO} + \left(\frac{32}{60}\right) X_{DRI}^{SiO_2} + \left(\frac{48}{102}\right) X_{DRI}^{Al_2O_3}\right] W_{DRI} + W_{O_2} = X_{CO}^{O} W_{CO} + \left[\left(\frac{16}{71.85}\right) X_{DRI}^{FeO} + \left(\frac{32}{60}\right) X_{DRI}^{SiO_2} + \left(\frac{48}{102}\right) X_{DRI}^{Al_2O_3}\right] W_{Slag}$
CO_2 balance: Ash content in lime and volatile matter in the coke
$X_{Cal}^{PxC} W_{Cal} + X_{Coke}^{M.V.} W_{Coke} = W_{CO_2}$

(continued)

Table 2.20 (continued)

Impurity restriction: all impurities, including residual elements. This example; 0.4%
Carbon requirements: Minimum carbon requirements $X^{C}_{DRI}W_{DRI} + X^{C}_{Scrap}W_{Scrap} + X^{C}_{Coke}W_{Coke} \geq X^{C}_{Steel}W_{Steel} + X^{FeO}_{DRI}\left(\frac{12}{71.85}\right)W_{DRI}$
Total mass balance
$W_{DRI} + W_{Scrap} + W_{Cal} + W_{O_2} + W_{Coke} = W_{Slag} + W_{CO} + W_{CO_2} + W_{Steel} + W_{Fume}$
Total energy balance
$\Delta H_{DRI}W_{DRI} + \Delta H_{Scrap}W_{Scrap} + \Delta H_{Coke}W_{Coke} + \Delta H_{Cal}W_{Cal} + \Delta H_{O_2}W_{O_2}$ $-\Delta H_{Fume}W_{Fume} + \Delta H_{R_{Endo}} - \Delta H_{R_{Exo}} + (1 - \text{Efic.})\left(\sum \Delta H_{R_{Exo}}\right) = [1 - (1 - \text{Efic.})]0.83\ \text{E.E.}$
Objective function: minimization
$C = C_{DRI}W_{DRI} + C_{Scrap}W_{Scrap} + C_{Coke}W_{Coke} + C_{Cal}W_{Cal} + C_{O_2}W_{O_2} + C_{E.E}.\text{E.E.}$

Where: X_{ij}, mass fraction specie i in material j; W_{slag}, slag weight; W_A, alloys weight; W_{Scrap}, scrap weight; cal, fluxes; E.E., electric energy; C_i, materials cost or energy per ton of steel; ΔH_i, sensible heat, reaction heat and energy losses

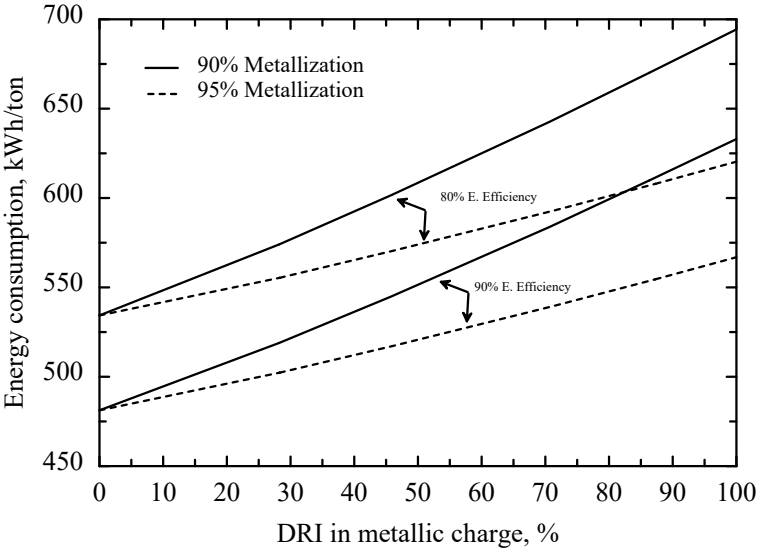


Fig. 2.7 Effect of % DRI on electric energy consumption, as a function of metallization of DRI and thermal efficiency [15, 44]

metallic charge changes from 100% scrap to 100% DRI with 90% metallization and 90% thermal efficiency. If the metallization is improved to 95% the increment on energy consumption is reduced by half to 80 kWh/ton, in other words the impact factor ranges from about 0.8 to 1.5 kWh/ton for every 1% of DRI. In practice a value of 1 kWh/ton can be applied.

The influence of metallization and carbon content in DRI on energy consumption is shown in Fig. 2.8. The impact factors of metallization and carbon content on energy consumption are 11 and 32 kWh/ton, respectively, for every increment of 1% in each variable. This result indicates a stronger effect of the carbon content in DRI.

Figure 2.9 shows the effect of carbon in DRI for different degrees of metallization on the consumption of electric energy consumption. Carbon has a strong effect to decrease energy consumption because is a source of chemical energy. Before the 1990s, the typical carbon concentration in DRI was about 1%. Suppliers of DRI technology had the idea that DRI should be produced with as much metallic iron units and possible, in this way increasing carbon would dilute the concentration of iron and was against the production of value. Progressively from the 1990s with the interest to decrease the consumption of electric energy and replace it by chemical energy there was a full change in philosophy, it became accepted the need to increase the concentration of carbon to higher values, in the range from 2 to 4%C. The optimization model suggests that an increment of 1%C in DRI decreases energy consumption by about 32 kWh/ton.

Figure 2.10 shows the effect of oxygen injection on energy consumption. The model predictions indicate that for each 1 Nm^3 of oxygen per ton of steel there is

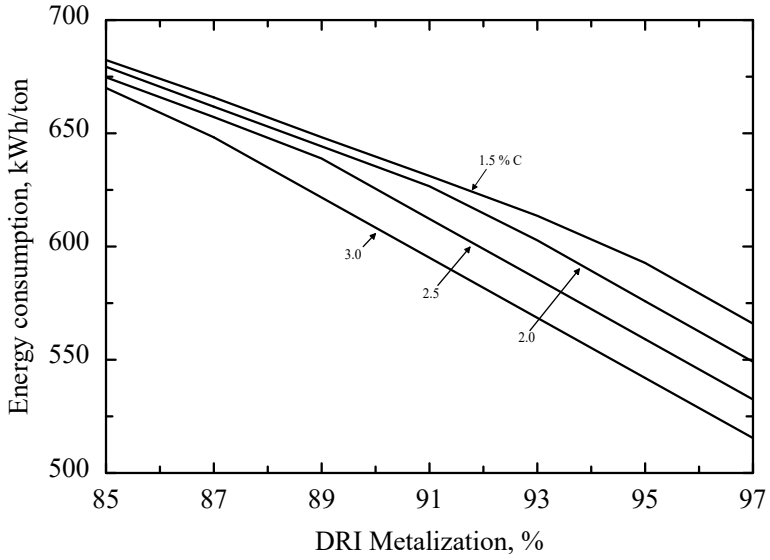


Fig. 2.8 Effect of metallization and carbon content of DRI on electric energy consumption [15, 44]

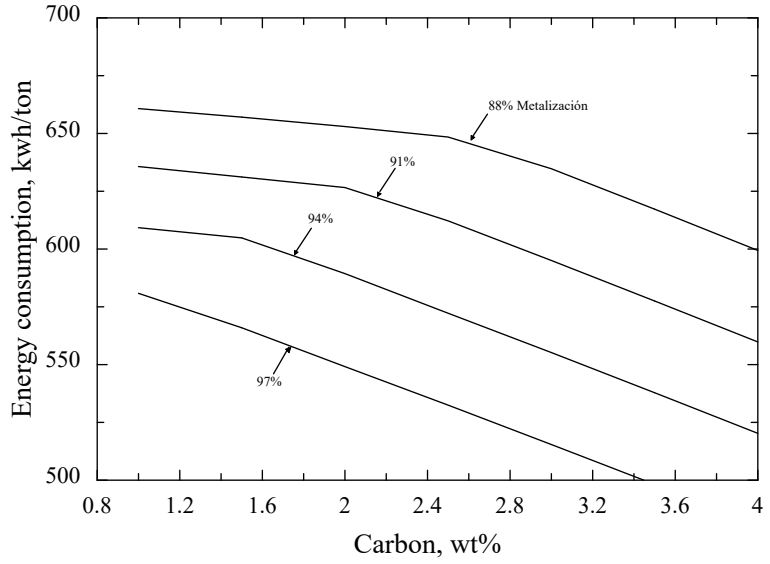


Fig. 2.9 Effect of carbon in DRI on energy consumption at different degrees of metallization [15, 44]

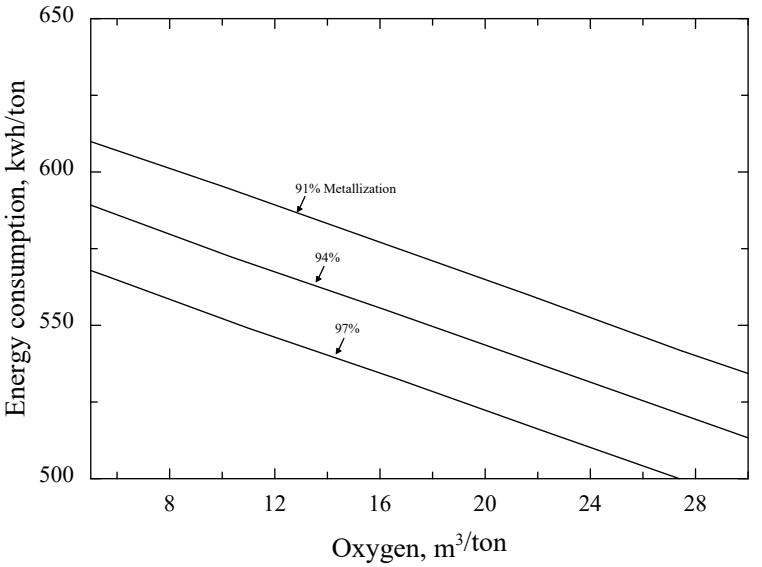


Fig. 2.10 Effect of oxygen injection on energy consumption at different degrees of metallization [15, 44]

a decrease in energy consumption of 3 kWh/ton. Oxygen injection is described in detail in Chap. 4.

Figure 2.11 shows the negative effect of the gangue content in DRI on energy consumption. The gangue content in DRI is one of the most critical aspects on DRI quality, in particular the amount of acid gangue, because its effects are multiplied. Its first negative effect is the energy for melting high melting point oxides and second, the additional energy to melt the additional fluxes required to control the slag chemistry. The model predictions indicate that increasing 1% gangue in DRI increases energy consumption by 15 kWh/ton.

Figure 2.12 reports Sankey diagrams describing total energy input and outputs using 100% DRI of 90 and 95% metallization and a thermal efficiency of 55%. The input of chemical energy is in the order of 20–25%. The heat losses in the off-gas are underestimated. CO_2 is produced only from volatile matter in the coke and calcination losses in lime, which explains the predicted low value of the off-gases and its sensible heat underestimated, from 10 to 12%.

It should be noticed that the impact factors obtained with this model result from the variation in the variables involved, in other words specific for the plant where the data was taken, however the model offers the potential to make a good sensitivity analysis of each variable, allowing to define its impact on both the production costs and energy consumption.

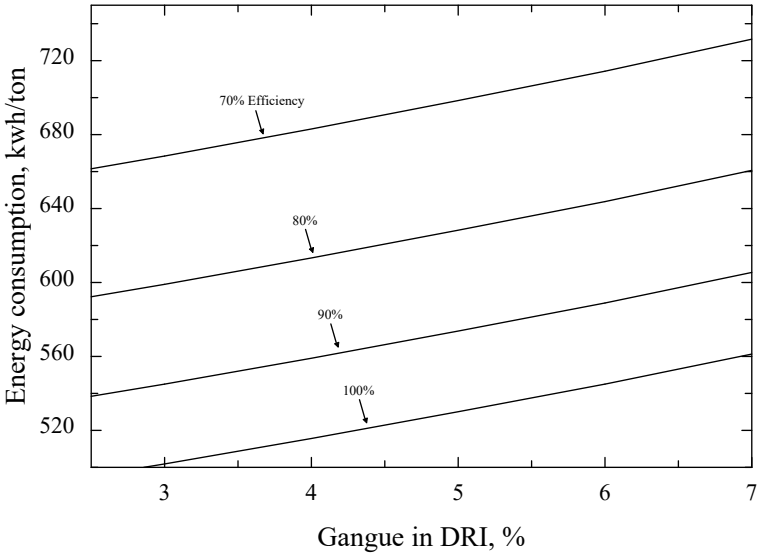


Fig. 2.11 Effect of gangue content in DRI on energy consumption at different thermal efficiencies [15, 44]

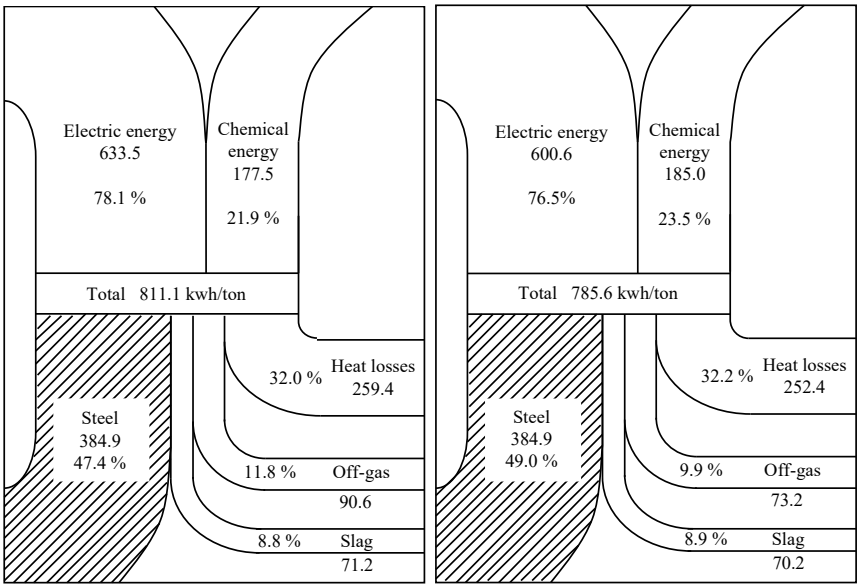


Fig. 2.12 Sankey diagrams indicating total inputs and outputs of energy for DRI of different metallization; 90% (left) and 95% (right) [15, 44]